Implied Volatility

What the number means, how it is computed, how to look at it, and what practitioners do with it

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Abstract

Implied volatility is neither a forecast nor a price. It is a unit: the volatility input that makes a standard pricing formula match an option's market price. This chapter explains the different true statements that one number makes, and how to compute and use it correctly.

Using the Black–Scholes formula to quote options does not mean believing it, in the same way that quoting a bond by yield does not mean believing in one constant rate. The unit removes the share price, strike, expiry and interest rate, so options become comparable. Each strike and expiry has its own implied volatility, and a single quoted figure is a convention. The number has three valid readings. It is a price in comparable units. It is the break-even daily move for a delta-hedged option: about 1.26% a day at 20% implied volatility. Across strikes, the pattern of implied volatilities encodes the risk-neutral distribution of the underlying. Two rules of thumb help. Dividing annual volatility by 16 gives the typical daily move. An at-the-money straddle costs about the expected size of the move over its life.

Implied volatility predicts realized volatility better than historical forecasts do, but it runs high. On the S&P 500 it has averaged about four points above subsequent realized volatility since 1990. The gap is the premium for selling insurance with small steady gains and rare large losses. Size short-volatility positions by the cost of the bad day. Probabilities read from option prices are risk-neutral. They overweight bad outcomes and give upper bounds on real-world probabilities. They come from the slope of the smile across strikes. In the worked example, plugging one strike's own volatility into the formula overstated the probability of a 10% fall by three quarters.

The VIX applies the same idea with different weights. A strip of out-of-the-money options replicates a variance swap, and discretizing that strip gives the VIX. The same view explains the drop in implied volatility after earnings: the event adds a fixed amount of variance to the term structure, and that variance is gone once the event passes. The chapter concludes that most practical difficulty is in the data. A forward price error of half a percent creates 4.4 points of false skew. Far out-of-the-money quotes give unreliable volatilities. The time-to-expiry convention alone can change an intraday estimate by a factor of 2.3. When a surface looks wrong, check the forward first.

What this is. A from-first-principles tutorial on implied volatility — the most quoted number in options markets, and one of the most casually misread. It gets called the market's forecast of volatility, the price of an option, the fear gauge, the width of a probability distribution and the break-even of a hedge. Each description is right about something and wrong about something else. This document takes the readings one at a time, shows exactly where each one comes from, and shows that they are faces of a single object. It then covers how the number is computed from real quotes (where nearly all of the practical difficulty lives), how it is drawn, and what market makers, volatility traders, risk managers, allocators and researchers actually do with it.

Who it is for. A reader who is mathematically comfortable and technically strong, but not an options specialist, and who intends to *use* implied volatility: to build a volatility surface, compute a VIX-style index, compare implied with realised volatility, read a probability off option prices, or feed option-implied quantities into a model. Intuition comes first throughout. Every formula is preceded by what it means and followed by a number. Basic option vocabulary is defined briefly in §2; Dealer Hedging and Gamma Exposure builds it from zero, and this document assumes you can look there when a definition here feels thin.

How to read it. Five parts, following the four questions in the title.

- **Part I (§1–§2) — the concept.** What implied volatility is, the intuitions to discard, the one idea that organises everything else, and just enough Black–Scholes to make the definition precise.
- **Part II (§3–§8) — ways to read the number.** A price in volatility units; the break-even rate of a hedge; the price of future variance; a forecast with a risk premium attached; and a probability distribution written in lognormal units. §8 joins the last two through local volatility and describes how the whole surface moves when the market moves.
- **Part III (§9–§11) — calculating it.** Inverting one price, building a surface from noisy quotes, and computing the derived quantities people actually quote: at-the-money volatility, skew, term structure, VIX-style variance, densities, event moves and implied correlation.

- **Part IV (§12) — visualising it.** The standard pictures, which question each one answers, and how each one misleads.
- **Part V (§13–§15) — using it.** How each kind of practitioner uses the number, a consolidated list of failure modes, and a synthesis.

§16 is the reference list, grouped by kind. Appendix A collects every concept the main text leans on without fully explaining, ordered so that it reads as a build-up.

If you read four things, read **§1.4** (the spine: a price, a rate and a shape), **§4.3** (why implied variance is a gamma-weighted average of the variance priced in), **§7.2–§7.4** (why a smile is a distribution, and the probability trap that follows), and **§9.4** (the inputs that matter far more than the root-finder).

Relationship to the other notes. Dealer Hedging and Gamma Exposure derives the Greeks and the hedged-P&L identity in detail (its sections 3 and 4) and explains who holds options and why that shows up in their prices; this document uses those results and refers back rather than repeating them. Simple and Log Returns explains why volatility is defined on log returns. Market Regimes and Hidden Markov Models is the right frame for using the VIX or the volatility term structure as a regime variable (§13.5). Stochastic Processes covers the Brownian motion the models here are built on. Each note stands alone.

Epistemic tags. Claims are flagged by status where the status changes what you should do:

- **[Fact]** — replicated across independent datasets or implementations, with broad agreement among people who have looked.
- **[Contested]** — documented, but with live disagreement about magnitude, robustness, or cause.
- **[Hypothesis]** — a proposed mechanism, not decisively tested.
- **[Practice]** — practitioner convention. May well be right; the evidence is private or absent.

Untagged sentences are definitions, derivations, or arithmetic. Numbers that come from simulations or stylised surfaces I built while writing are labelled **[Simulated]**; the generating code is committed alongside this document in `figures/iv_*.py`, and each script prints the numbers quoted in the text, so you can change the parameters and rerun.

Notation. The **underlying** is the asset an option is written on; S (or S_t) is its price. An option has **strike** K and **expiry** T; $\tau = T - t$ is the time remaining, in years, counted in calendar days over 365 unless a passage says otherwise. r is the risk-free rate and q the underlying's dividend (or carry) yield. $F = Se^{(r-q)\tau}$ is the **forward price** for the option's expiry and $D = e^{-r\tau}$ the **discount factor**. $k = \ln(K/F)$ is **log-moneyness**: negative for strikes below the forward, positive above.

V is an option's value, C a call and P a put. $N(\cdot)$ is the standard normal cumulative distribution function and $\varphi(\cdot)$ its density. $\mathrm{BS}(\cdot)$ denotes the Black–Scholes price as a function of its inputs, and d_1, d_2 are the quantities of §2.2. The Greeks are $\Delta = \partial V/\partial S$, $\Gamma = \partial^2 V/\partial S^2$, $\nu = \partial V/\partial \sigma$ (vega) and $\Theta = \partial V/\partial t$; ΓS^2 is **dollar gamma**.

Several volatilities must be kept apart, and the document is careful about which is meant:

Symbol	Name	What it is
σ_i, or $\sigma_i(K,T)$	implied volatility	The number that makes BS reproduce a market price (§1.1); as a function of strike and expiry, the **surface**
σ_t	instantaneous volatility	The true, unobserved volatility of the underlying at time t in whatever process actually drives it
σ_r	realised volatility	The volatility a price path actually delivered, measured afterwards from returns
$\sigma_{\mathrm{loc}}(S,t)$	local volatility	The volatility as a function of price and time that reproduces the whole surface (§8)
σ_{VS}	variance-swap volatility	The square root of the fair variance of §5, of which the VIX is an instance

$w(k,T) = \sigma_i^2(k,T)\,T$ is **total implied variance** at log-moneyness k and expiry T — the variance of the log return to expiry that the option price implies. Volatility is quoted in percent; a **vol point** is one percentage point of σ. A surface is always described as seen today, so an expiry T and the time remaining to it coincide; $w(k,T)$ and $\partial w/\partial T$ follow the literature in writing T.

Two probability measures appear. $\mathbb{Q}$ is the **pricing** (risk-neutral) measure under which discounted prices are expectations, and $\mathbb{P}$ is the **physical** (real-world) measure under which events actually happen; the blackboard letters keep $\mathbb{P}$ apart from the put price P. $f_T(K)$ is the risk-neutral density of S_T at K. Π is the profit and loss (P&L) of a hedged position. ρ denotes a correlation — between price and volatility changes, or between stocks in an index (§11.7) — or the rotation parameter of the SVI smile (§10.4), which plays the same role; the passage always says which.

The running example is shared with the Dealer Hedging note: an underlying at 100, a strike of 100, 30 calendar days to expiry ($\tau = 30/365$), $r = q = 0$ (so $F = S = 100$ and $D = 1$), and 20% volatility. The call and the put each cost **2.29**.

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- A. Concepts and prerequisites — every idea the main text leans on without fully explaining, built from first principles and ordered by dependency.

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1. The number, and the intuitions to discard

1.1 The definition, with one number

An option's price depends on six inputs: the underlying price S, the strike K, the time to expiry τ, the interest rate r, the dividend yield q, and the volatility σ of the underlying over the option's life. Five of them can be looked up. The sixth cannot, because it is a property of the future.

The Black–Scholes formula (§2.2) turns all six into a price. Implied volatility runs the formula the other way: take the price the market is actually paying, hold the five observable inputs fixed, and find the volatility that makes the formula agree.

$$\mathrm{BS}(S, K, \tau, r, q, \sigma_i) = V_{\text{market}}.$$

The solution σ_i is the option's **implied volatility**. For the running example, a 30-day at-the-money call trading at 2.29 has an implied volatility of 20%. If buyers bid it up to 2.52, its implied volatility becomes 22%: the option's vega is 0.114 per vol point, so 23 cents of extra premium is two vol points.

Three things about this definition matter for everything that follows.

- **The formula is used as a conversion table, not as a description of the world.** Nothing in the definition says anyone believes returns are lognormal or that volatility is constant. The formula is a ruler: a fixed, agreed way of turning a dollar price into a number that can be compared across options.
- **It works only because price rises with volatility.** Every option is worth more when volatility is higher (§2.3), so each price corresponds to exactly one volatility. That monotonicity, not any belief about markets, is what makes the number well defined.
- **One option gives one number; a chain of options gives a surface.** Every strike and every expiry has its own implied volatility, and they are generally all different. "The implied volatility" of a stock or an index is always shorthand for one point on that surface, chosen by convention (§11.1).

1.2 An exact analogy: the yield of a bond

A bond's **yield to maturity** is the single discount rate that makes the present value of its promised cash flows equal its market price. The calculation assumes a flat interest-rate curve, which nobody believes. Bonds are quoted and compared in yield anyway, because the conversion strips out coupon size, maturity and face value and leaves a number that means roughly the same thing for every bond. When yields differ across maturities, nobody concludes that the yield formula is broken; they draw a **yield curve** and read it as information about what the flat-rate assumption leaves out — expected future rates, and a term premium for bearing duration risk.

Implied volatility is the same construction applied to options:

	Bonds	Options
Observed	Price	Price
Ruler (a model nobody believes literally)	Flat-rate discounting	Black–Scholes with constant volatility
The number it yields	Yield to maturity	Implied volatility
Why the ruler works	Price falls monotonically as yield rises	Price rises monotonically with volatility
Variation across instruments	Yield curve, by maturity	Volatility surface, by strike and expiry
What the variation measures	Expected rates plus term premium	Expected variance plus variance risk premium, and the shape of the distribution

The analogy is exact in structure, and it survives one level deeper: a yield contains a risk premium on top of expected future rates, and an implied volatility contains a risk premium on top of expected future variance (§6). Riccardo Rebonato's often-repeated description of implied volatility is that it is the wrong number to put into the wrong formula in order to obtain the right price (Rebonato, 1999). The joke works because both wrongs are deliberate and cancel exactly.

Where the analogy stops: a bond's cash flows are fixed, so its yield is a statement about discounting alone. An option's payoff depends on the path, and in particular on how much variance the path delivers where the option is sensitive to it. That is why implied volatility has more readings than yield does.

1.3 Seven intuitions to discard

Most confusion about implied volatility comes from a handful of intuitions that are half right. Naming them now saves a great deal of unlearning later.

The intuition	Why it fails	What is true instead	Section
"Implied volatility is the market's forecast of volatility"	It is a price, and prices contain risk premia. S&P 500 implied volatility has exceeded subsequently realised volatility by about 4 vol points on average since 1990	A forecast plus a premium, plus the effect of supply and demand for options	§6
"It is a parameter of a model the market believes"	The model is known to be false, and markets use it anyway	A unit of measurement; the *shape* of the surface records how the model is false	§3, §7
"A stock has an implied volatility"	Every strike and expiry has its own	A surface; any single number is a convention	§10, §11.1
"High implied volatility means options are expensive"	Expensive relative to what?	Rich or cheap only relative to the variance the option will actually experience where it has gamma, net of the premium for bearing that risk	§4, §6
"An option's implied volatility is the underlying's volatility over the option's life"	Variance that arrives far from the strike barely affects an option	A dollar-gamma-weighted average of the variance priced along the paths from today's price to the strike	§4.3
"Implied volatility measures moves in either direction equally"	Out-of-the-money puts and calls on an index trade at very different implied volatilities	Different strikes price different parts of the distribution; the downside wing prices crash risk	§7
"$N(d_2)$ at the option's implied volatility is the probability of finishing in the money"	It ignores the skew, and it is the wrong probability measure	With a typical index skew, the naive number overstates the chance of finishing below a 10%-lower strike by about three quarters; and it is a pricing probability, not a real-world one	§6.1, §7.4

1.4 The spine: a price, a rate, and a shape

Three statements organise this whole document. Each later section is one of them viewed up close, and the synthesis in §15 is the three of them viewed together.

1. Implied volatility is a price. It is an option price re-expressed in the unit in which a lognormal, constant-volatility world would be flat. That is the definition (§1.1), and it is why options are quoted, compared and marked in volatility (§3).

2. Implied volatility is a rate. The P&L of a delta-hedged option over a short interval is half its dollar gamma times the gap between the variance that actually happened and the variance implied volatility charged for:

$$d\Pi = \tfrac{1}{2}\,\Gamma S^2\left(\sigma_t^2 - \sigma_i^2\right) dt.$$

So implied variance is the **break-even variance rate** for anyone who hedges the option (§4). Take expectations under the pricing measure, and demand that a fairly priced option break even on average: implied variance becomes a **weighted average of the variance the market has priced in**, with weights equal to the option's dollar gamma along the paths it might follow. This is the master equation of the document, derived in §4.3:

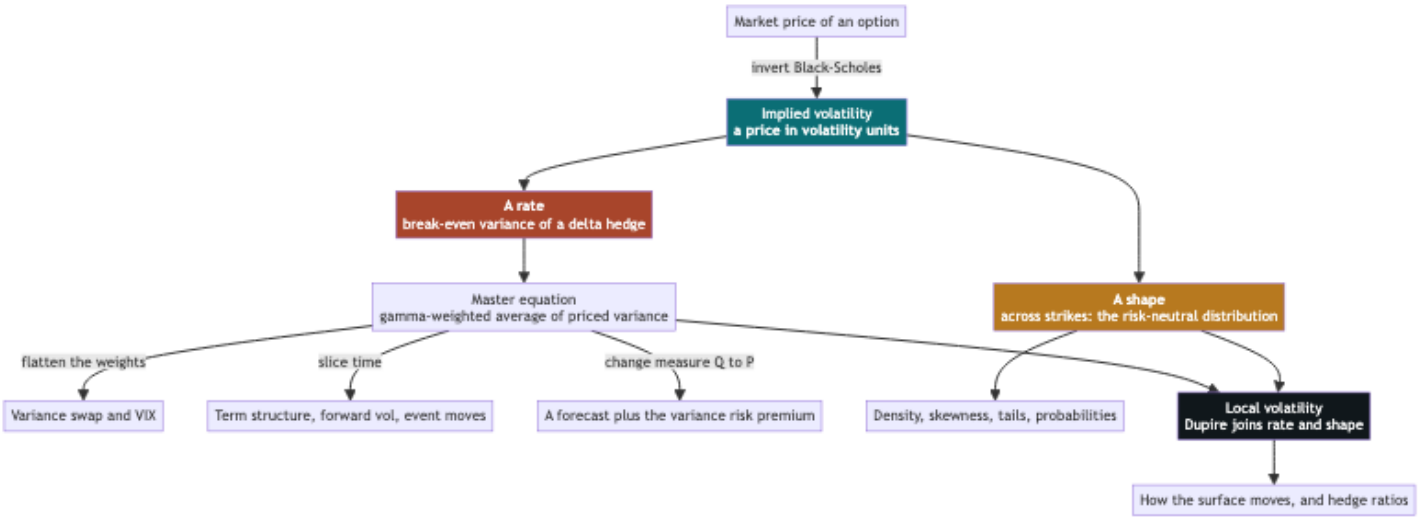
$$\sigma_i^2 \;=\; \frac{\mathbb{E}^{\mathbb{Q}}\left[\displaystyle\int_0^T \Gamma_t S_t^2\,\sigma_t^2\,dt\right]}{\mathbb{E}^{\mathbb{Q}}\left[\displaystyle\int_0^T \Gamma_t S_t^2\,dt\right]}.$$

Three knobs in this equation generate most of the named readings of implied volatility. Change the **weights** — replace one option's dollar gamma with the flat dollar gamma of a strip of options weighted by $1/K^2$ — and you get the variance swap

and the VIX (§5). Change the **measure** from $\mathbb{Q}$ to $\mathbb{P}$ and you get a forecast, with the variance risk premium as the gap (§6). Change the **slice of time** the integral covers and you get forward volatility and the variance of a single event, because total variance adds up over time (§5.4–§5.5).

3. Implied volatility is a shape. Across strikes at one expiry, option prices pin down the whole risk-neutral probability distribution of the price at expiry: the second derivative of the call price with respect to the strike *is* the density (§7.2). The smile is that distribution written relative to a lognormal. Its level is the width of the distribution, its slope is the skewness, and its curvature is the fatness of the tails (§7.3).

The rate and the shape are joined by **local volatility** (§8). Dupire's formula says the local variance at a strike and expiry is the rate at which total implied variance grows with expiry, divided by a term that is proportional to the risk-neutral density at that strike — a time-direction quantity over a strike-direction quantity. The two no-arbitrage conditions every surface must satisfy are exactly the conditions that the numerator be non-negative and the denominator positive.



The rest of Part II is a tour of this diagram, one reading per section, all using the same template: the intuition, the formal statement, a number, what the reading is good for, and how it misleads.

Reading	What changes in the master equation	Section	What it is good for
A price in volatility units	Nothing: this is the definition	§3	Quoting, comparing and marking options
The break-even rate of a hedge	Weights are this option's dollar gamma	§4	Deciding whether to own or sell an option you will hedge
The price of variance	Weights flattened by a $1/K^2$ strip; time sliced	§5	The VIX, variance swaps, term structure, event moves
A forecast with a premium	Measure $\mathbb{Q}$ replaced by $\mathbb{P}$	§6	Volatility forecasting; the variance risk premium
A distribution	Vary the strike instead of integrating over time	§7	Probabilities, tails, skew
An average of local volatility	Weights live on a bridge from spot to strike	§8	Pricing and hedging beyond vanilla options; surface dynamics

1.5 How the idea evolved

Year	Development	What changed
1973	Black–Scholes and Merton; listed equity options begin trading in Chicago	An option has a price, and one input that cannot be observed
1976	Implied standard deviations computed from option prices; Black's formula for options on futures	The number gets its name, and its first test as a forecast
1987	The October crash	The index skew appears, and never leaves
1993	Cboe's first volatility index, on S&P 100 options; the Heston model	Volatility becomes something you can quote, and something you can model as a process
1994	Local volatility, by Dupire and by Derman and Kani	One diffusion is shown to fit an entire surface
1999	Variance swaps and the replicating strip	Volatility becomes tradable without delta hedging
2002	SABR	A smile model built for hedging rather than only for fitting
2003	The VIX is redefined as model-free variance on S&P 500 options	The index becomes a variance-swap rate
2004	VIX futures list; the SVI parameterisation appears	Volatility becomes an asset class with its own term structure
2009	The variance risk premium is shown to predict equity returns	The gap between implied and realised becomes a research object
2014	Arbitrage-free SVI	Surface fitting acquires explicit no-arbitrage conditions
2018	"Volatility is rough"	The shape of the short end becomes a research field
2022	SPX options expire every trading day	Intraday implied volatility, and the clock behind it, become first-order

The history falls into four eras, each with a one-line thesis.

1973–1986: implied volatility as a by-product. Black and Scholes (1973) and Merton (1973) gave a price that depended on one unobservable input, and within three years Latané and Rendleman (1976) were computing the "implied standard deviation" from option prices and asking whether it forecast future volatility. What changed: volatility became something you could read off a market rather than only estimate from history. The limitation, fairly judged, was that the number was treated as a single estimate of a single parameter. Index smiles were mild before 1987, so little seemed lost. What survives: the definition, unchanged.

1987–1999: the smile becomes permanent, and gets modelled. After the October 1987 crash, S&P 500 options have priced downside strikes at persistently higher implied volatility than upside strikes (Rubinstein, 1994). [Fact] Practitioners stopped treating the smile as noise and started treating it as information. Heston (1993) produced smiles from stochastic volatility; Dupire (1994) and Derman and Kani (1994) showed how to recover the unique local volatility function consistent with the whole surface. At the end of the decade, Carr and Madan (1998) and Demeterfi, Derman, Kamal and Zou (1999) showed that a strip of options replicates a contract on realised variance, making volatility itself tradable without any delta hedging. The limitation: models that fit today's surface often predict tomorrow's surface badly (§8.5). What survives: all of it, as the standard toolkit.

2000–2014: implied volatility as an index, an asset and a risk factor. In 2003 Cboe redefined the VIX as a model-free variance computed from S&P 500 options (Carr and Wu, 2006), and futures and options on it followed. Academic work measured the gap between implied and realised variance as a risk premium (Carr and Wu, 2009) and found that it predicts equity returns (Bollerslev, Tauchen and Zhou, 2009). Practitioners standardised how surfaces are parameterised, notably with SVI, later given explicit no-arbitrage conditions by Gatheral and Jacquier (2014). The limitation: volatility products concentrated short-volatility risk in ways that were poorly understood until they failed (February 2018; Dealer Hedging, section 8.4). What survives: the VIX as the world's default measure of equity risk appetite.

2015 onward: every horizon, every hour. Options now expire daily, implied volatility is computed intraday for options with hours to live, and the shape of the short end of the surface has become a research field of its own — including the [Contested] claim that the at-the-money skew follows a power law in maturity because volatility is "rough" (Gatheral, Jaisson and

Rosenbaum, 2018; §8.4). What changed: the clock used to measure time to expiry, a detail for 30-day options, became first-order (§9.4).

§1 Key takeaways

1. Implied volatility is the volatility that makes the Black–Scholes formula reproduce a market price. The formula is used as a ruler, not as a belief.
2. The number is well defined only because option prices rise monotonically with volatility.
3. It is exactly analogous to a bond's yield: a price converted into a comparable rate through a deliberately simple model, whose variation across instruments is information.
4. Every strike and expiry has its own implied volatility. "The" implied volatility of an asset is a convention.
5. Implied volatility is simultaneously a price, a rate and a shape: the price of the option in volatility units, the break-even variance rate of a delta hedge, and — across strikes — the risk-neutral distribution written relative to a lognormal.
6. Implied variance is a dollar-gamma-weighted average of the variance priced in. Changing the weights gives the variance swap and the VIX; changing the measure gives a forecast plus a risk premium; slicing time gives forward volatility.
7. It is not a pure forecast, not one number, not symmetric, and not a source of real-world probabilities without correction.

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2. The ruler: just enough Black–Scholes

2.1 Volatility, and the square root of time

Volatility σ is the annualised standard deviation of log returns. If daily log returns are independent, variances add, so the standard deviation over n days grows like $\sqrt{n}$. Over an option's remaining life τ the standard deviation of the log return is $\sigma\sqrt{\tau}$, which this document calls **total volatility**; its square $\sigma^2\tau$ is **total variance**.

With about 252 trading days a year and $\sqrt{252} \approx 16$, traders use the **rule of 16**: 16% annual volatility is roughly a 1% move on a typical trading day. The running example's 20% is a typical daily move of 1.26% on trading days (or 1.05% if the year is counted as 365 calendar days, which spreads the same variance over more days), and a standard deviation of $20\% \times \sqrt{30/365} = 5.73\%$ over the option's 30 days. Simple and Log Returns explains why log returns, not simple returns, are the natural unit here.

2.2 What an option price really depends on

The Black–Scholes price of a European call, written in terms of the forward F and discount factor D (the "Black" form, Black, 1976a), is

$$C = D\left[F\,N(d_1) - K\,N(d_2)\right], \qquad d_{1,2} = \frac{\ln(F/K) \pm \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}},$$

and the put follows from **put–call parity**, $C - P = D(F - K)$, which holds for European options regardless of any model. Read the formula as a recipe: $N(d_2)$ is the risk-neutral probability that the call finishes in the money, and d_2 is the distance from the forward to the strike measured in standard deviations of the log return (with a small adjustment of half a variance). Dealer Hedging, sections 3.2 and 3.3, derives the formula from replication.

Now divide by DF and write the strike as log-moneyness $k = \ln(K/F)$:

$$\frac{C}{DF} = N(d_1) - e^k N(d_2), \qquad d_{1,2} = -\frac{k}{\sigma\sqrt{\tau}} \pm \frac{\sigma\sqrt{\tau}}{2}.$$

The normalised price depends on **exactly two numbers**: how far the strike is from the forward, k, and how much the log price is expected to spread before expiry, $\sigma\sqrt{\tau}$. Three consequences run through the rest of the document.

- **Rates and dividends enter only through F and D.** Getting the forward right is therefore most of what "getting the inputs right" means (§9.4).
- **Volatility and time enter only through $\sigma\sqrt{\tau}$.** A one-year option at 10% volatility and a four-year option at 5% volatility with the same moneyness have the same normalised price. What the option really prices is total volatility; the annualised σ is a convention for comparing across expiries.
- **Total implied variance $w = \sigma_i^2\tau$ is the natural coordinate.** It is what the price actually pins down, and it adds up over time (§5.4). Surfaces are best built, interpolated and checked for arbitrage in (k, w) rather than in (K, σ_i) (§10).

Put–call parity has a consequence that is easy to state and powerful in practice: a call and a put with the same strike and expiry must have **the same implied volatility**, because the same σ that prices one prices the other. When computed call and put volatilities disagree, something is wrong with the inputs — usually the forward — or the options are American (§9.4).

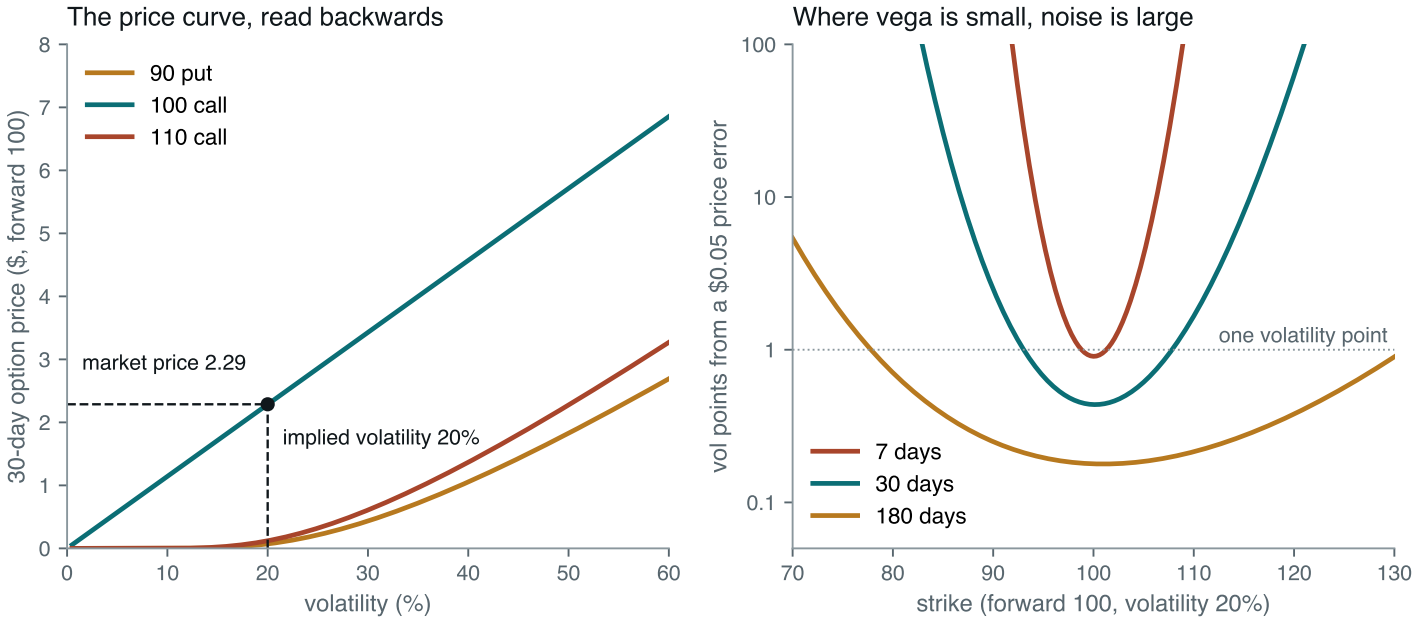
2.3 Why the formula can be run backwards

Vega, the sensitivity of the price to volatility, is

$$\nu = \frac{\partial C}{\partial \sigma} = DF\,\varphi(d_1)\sqrt{\tau} \;>\; 0,$$

strictly positive for any strike and any positive time to expiry, and the same for a call and a put. The intuition is convexity: an option pays off asymmetrically, so spreading out the distribution of the final price adds more to the winning outcomes than it takes from the losing ones.

Positive vega, together with the limits of the price as volatility goes to zero and to infinity, is what makes implied volatility well defined. As $\sigma \to 0$ the call is worth its discounted intrinsic value against the forward, $D(F-K)^+$; as $\sigma \to \infty$ it is worth DF. Every price strictly between those bounds corresponds to **exactly one** volatility. A quoted price outside them has no implied volatility at all — and since those bounds are no-arbitrage bounds, such a price is either an arbitrage or, far more often, bad data.



The left panel is the whole idea of implied volatility in one picture: each curve is monotone, so a price on the vertical axis can be read back to a volatility on the horizontal axis. Two features of the curves matter later. The at-the-money curve is almost a straight line through the origin (§2.4), so its inversion is easy and robust. The out-of-the-money curves are flat near zero volatility and rise steeply only later; where a curve is flat, a small change in price corresponds to a large change in volatility. The right panel measures that fragility, and §9.3 returns to it.

2.4 The at-the-money shortcut, and the straddle as an expected move

At the money forward, $K = F$, so $k = 0$ and $d_{1,2} = \pm\frac{1}{2}\sigma\sqrt{\tau}$. Expanding the normal distribution function around zero,

$$C_{\text{ATM}} = DF\,[N(\tfrac{1}{2}\sigma\sqrt{\tau}) - N(-\tfrac{1}{2}\sigma\sqrt{\tau})] \;\approx\; DF\,\sigma\sqrt{\tau}\,\varphi(0) \;=\; \frac{DF\,\sigma\sqrt{\tau}}{\sqrt{2\pi}} \;\approx\; 0.4\,DF\,\sigma\sqrt{\tau}.$$

This is the approximation of Brenner and Subrahmanyam (1988), and it is remarkably accurate for the options people trade most. For the running example it gives $0.4 \times 100 \times 0.20 \times 0.2867 = 2.293$ against an exact 2.287. Run it backwards and a trader can read implied volatility off an at-the-money price without a computer: $\sigma_i \approx C/(0.4\,DF\sqrt{\tau}) = 19.94\%$. The shortcut's real lesson is that the at-the-money price is **almost linear in volatility**, which is why at-the-money implied volatility is the most robust number on the surface and why at-the-money vega barely depends on the level of volatility.

The **straddle** — a call plus a put at the same strike — has a second reading that practitioners use daily. At the money it costs about twice the call, $0.8\,DF\sigma\sqrt{\tau}$. And if the change in the forward over the option's life is roughly normal with standard deviation $F\sigma\sqrt{\tau}$, the expected *absolute* change is $F\sigma\sqrt{\tau}\sqrt{2/\pi} = 0.798\,F\sigma\sqrt{\tau}$. The two coincide:

> **An at-the-money straddle costs the discounted expected absolute move of the underlying over the option's life** — under the pricing measure, and to the extent the move is close to normal.

The running example's straddle costs 4.57, which is 4.57% of the price: the market is pricing an average absolute move of about 4.6% over 30 days, corresponding to a standard deviation of $4.57/0.798 = 5.73\%$. This is the "expected move" quoted around earnings announcements (§5.5, §13.3). [Practice] Two caveats travel with it. It is a risk-neutral expectation, so it includes whatever premium sellers of options demand (§6). And fat tails change the ratio: for a distribution with more weight in the tails and more in the centre, the mean absolute move is a smaller fraction of the standard deviation than 0.8, so dividing a straddle price by 0.8 slightly understates the standard deviation the market is pricing.

2.5 Other rulers: normal volatility and variance units

Black–Scholes is not the only ruler in use, and the choice of ruler changes what the smile looks like.

- **Lognormal (Black–Scholes or Black-76) volatility**, in percent per square-root year. The convention for equities, foreign exchange and most commodities.
- **Normal (Bachelier) volatility**, in price units per square-root year — basis points a year for interest rates. The model lets the underlying's *change*, rather than its log change, be normally distributed. Near the money the two are related by $\sigma_N \approx \sigma_{\text{LN}} F$. Interest-rate options moved largely to normal-volatility quoting as rates approached zero, where lognormal volatility becomes enormous and unstable for small changes in the rate. [Practice]
- **Variance units.** Variance swaps are quoted as a volatility strike but pay on variance (§5.1), and some models are naturally stated in variance.

The ruler matters because a surface that is flat under one ruler is sloped under another. If a market prices options exactly as the normal model with $\sigma_N = 20$ price units a year on a forward of 100, the equivalent lognormal volatility at strike K is approximately $\sigma_N \ln(F/K)/(F-K)$ (Hagan, Kumar, Lesniewski and Woodward, 2002): 21.1% at a strike of 90 and 19.1% at 110. A "flat" normal smile is a lognormal skew of about one vol point per ten points of strike. Part of any skew is therefore a statement about which ruler is being used, and comparisons of skew across markets quoted under different conventions need conversion first.

§2 Key takeaways

1. Volatility is the annualised standard deviation of log returns; over an option's life the relevant number is total volatility $\sigma\sqrt{\tau}$. By the rule of 16, 16% is about 1% a trading day.
2. A normalised option price depends only on log-moneyness $k = \ln(K/F)$ and total volatility. Rates and dividends matter only through the forward and the discount factor.
3. Total implied variance $w = \sigma_i^2\tau$ is the coordinate the price actually pins down, and the right one for building surfaces.
4. Vega is strictly positive, so every price inside the no-arbitrage bounds has exactly one implied volatility; a price outside them has none.
5. A call and a put with the same strike and expiry have the same implied volatility. Disagreement diagnoses bad inputs.
6. At the money, price is almost linear in volatility: $C \approx 0.4\,DF\sigma\sqrt{\tau}$. The at-the-money straddle costs roughly the expected absolute move.
7. The ruler is a convention. Normal and lognormal volatilities give differently shaped smiles for the same prices.

3. A price in volatility units

3.1 What the conversion removes

The first reading of implied volatility is the literal one: it is the option's price, wearing different clothes. What the change of clothes accomplishes is best seen by looking at prices that have nothing in common and volatilities that are identical.

Option (20% volatility throughout)	Price
30-day at-the-money call, underlying at 100	2.29
30-day at-the-money call, underlying at 500	11.44
One-year at-the-money call, underlying at 100	7.97
30-day 110-strike call, underlying at 100	0.12
30-day 90-strike put, underlying at 100	0.07

Every option in that table is priced at exactly 20% volatility, and their prices span two orders of magnitude. Run it the other way and the point is sharper still: a price of **2.29** is 20% volatility for the 30-day at-the-money call, 41.5% for a *seven-day* at-the-money call, 5.7% for a *one-year* at-the-money call, and 50.2% for the 30-day 110-strike call. The dollar price on its own says nothing about whether an option is dear.

The conversion strips out four things at once: the level of the underlying, the strike, the time to expiry, and the financing and dividends (which enter only through the forward and the discount factor, §2.2). What is left is a number on a scale where "20%" means approximately the same thing for a 30-day option on a $40 stock and a two-year option on an index at 6,000.

3.2 Trading, quoting and marking in volatility

Because the number is comparable, the market uses it as the unit of account.

Quotes. Over-the-counter foreign-exchange options are quoted directly in volatility, with the strike specified by delta rather than by price (§10.4). Interdealer trades in listed index options are often negotiated the same way: the two sides agree a volatility and a reference price for the underlying, and the trade prints as a package of options plus the delta hedge. [Practice] The dollar price is then an output, computed from the agreed volatility.

Marks and P&L attribution. A dealer's book is marked to a fitted surface rather than to individual screen prices, and the day's P&L is attributed to the Greeks. Take the running example: the underlying rises 1%, one day passes, and implied volatility falls by one point. The call's value changes by $+0.397$, decomposed as

$$\underbrace{+\,0.511}_{\text{delta}} \;+\; \underbrace{+\,0.035}_{\text{gamma}} \;+\; \underbrace{-\,0.038}_{\text{theta}} \;+\; \underbrace{-\,0.114}_{\text{vega}} \;=\; +0.394,$$

with the remaining third of a cent coming from higher-order terms. The vega line — "the market repriced volatility and it cost us eleven cents" — exists only because prices have been converted into volatility. Without the conversion there is a P&L number and no explanation of it.

Relative value. Once every option on an underlying is a point on a surface, the natural questions become comparative: this put is 1.5 volatility points rich to the fitted skew; this expiry is cheap to the ones either side of it; this stock's volatility is high relative to its sector. Every one of those statements is a statement about residuals from a fitted surface (§10.4), which is why the fitting matters so much.

3.3 What the unit cannot remove

The conversion normalises the *model's* dimensions. It does not make different risks comparable, and four confusions follow from forgetting that.

- **Different underlying risks.** A 30% implied volatility on a biotech stock facing a binary regulatory decision and a 30% implied volatility on an equity index are not the same price for the same thing. The first is mostly the price of one event (§5.5); the second is the price of a diffusive process plus crash insurance.

- **Different parts of the distribution.** The implied volatility of a 25-delta put and that of an at-the-money option are prices of different risks, which is why they differ and why the difference is information rather than error (§7).
- **The risk premium travels with the number.** Comparing implied volatilities across underlyings compares risk premia and hedging demand as much as expected variance (§6.3). Single-stock options and index options carry systematically different premia (Driessen, Maenhout and Vilkov, 2009).
- **Model residue leaks in.** American exercise, discrete dividends, borrow costs and settlement conventions all distort the number if they are mishandled in the inversion (§9.4). A "high" implied volatility is sometimes just a wrong forward.

§3 Key takeaways

1. Implied volatility is the option's price in a unit that removes the underlying's level, the strike, the expiry, rates and dividends.
2. Dollar prices spanning two orders of magnitude can all be 20% volatility; one dollar price can be 5.7% or 50% volatility depending on the contract.
3. Markets quote, negotiate and mark in volatility, and P&L attribution's vega line exists only because of the conversion.
4. Relative-value statements about options are statements about residuals from a fitted surface.
5. The unit does not make different risks comparable: event risk, skew, risk premia and mishandled forwards all live inside the same number.

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4. The break-even rate of a hedge

4.1 The identity, in one paragraph

Buy an option, sell Δ shares against it, and keep the hedge current. Over a short interval in which the underlying moves by ΔS, the hedged position's P&L is, to second order,

$$\Delta\Pi \;\approx\; \Theta\,\Delta t + \tfrac{1}{2}\Gamma(\Delta S)^2,$$

because the delta terms cancel — that is what hedging is for. An option marked at implied volatility σ_i satisfies the Black–Scholes equation at that volatility, which with zero rates says $\Theta = -\tfrac{1}{2}\sigma_i^2 S^2\Gamma$: the option's time decay is exactly the rent charged for its gamma. Substituting,

$$\boxed{\Delta\Pi \;\approx\; \tfrac{1}{2}\,\Gamma S^2\left[\left(\frac{\Delta S}{S}\right)^2 - \sigma_i^2\,\Delta t\right]}$$

The P&L of a delta-hedged option is half its dollar gamma times the difference between the squared return that happened and the variance that implied volatility charged for. Dealer Hedging, section 4.4, derives this carefully, works it through a week of hedging, and simulates its distribution; this document takes it as given and asks what it says about the number σ_i.

It says that implied volatility is not really a forecast or an opinion. It is a **rate**: the rate at which a hedger pays (in time decay) for the right to collect squared returns, or collects in exchange for owing them.

4.2 The break-even move

Setting the bracket to zero gives the move at which a hedged option breaks even over an interval:

$$\left|\frac{\Delta S}{S}\right| = \sigma_i\sqrt{\Delta t}.$$

At 20% implied volatility, one day's break-even is 1.26% if the year is counted in 252 trading days, or 1.05% if it is counted in 365 calendar days — the same option, two conventions, and the difference matters over a weekend (§9.4).

Implied volatility	Daily break-even (252-day year)	Daily break-even (365-day year)
10%	0.63%	0.52%
16%	1.01%	0.84%
20%	1.26%	1.05%
32%	2.02%	1.68%
80%	5.04%	4.19%

This is the most practical single reading of implied volatility, and it converts the number into a question a trader can actually answer: *will this underlying move more or less than this each day?* Owning a hedged option is a bet on "more"; selling one is a bet on "less"; and because the payoff is quadratic in the move, the loser on a big day loses far more than the winner gains on a quiet one. At 20% implied volatility a 2% day delivers three and a half times the break-even variance, and a 3% day more than eight times.

Over the option's whole life, the expected P&L of selling at σ_i and living through realised volatility σ_r is the difference in Black–Scholes values, $V(\sigma_i) - V(\sigma_r)$, which is close to $\nu \times (\sigma_i - \sigma_r)$: about $\$1.14$ per option for each 10 volatility points, in the running example. But that is an *expectation.* The realised outcome around it is hedging error, it falls only as the square root of the number of rehedges, and it does not diversify across options on the same underlying (Dealer Hedging, section 4.7).

4.3 Implied variance is a gamma-weighted average

The identity holds interval by interval. Integrating it over the option's life turns implied volatility from a per-day break-even into a statement about the entire future, and this is the master equation of the document.

Work under the pricing measure $\mathbb{Q}$ with zero rates, and let the underlying follow *any* continuous process whose instantaneous volatility σ_t may itself be random: $dS_t = \sigma_t S_t \, dW_t$. Let $V(S, t; \sigma_i)$ be the Black–Scholes value of our option at the constant volatility σ_i. Itô's lemma plus the Black–Scholes equation at σ_i (the same substitution as in §4.1) gives

$$dV = \Delta \, dS + \tfrac{1}{2}\Gamma S^2 \left(\sigma_t^2 - \sigma_i^2\right) dt.$$

Integrate from now to expiry. At expiry the Black–Scholes value equals the payoff, so

$$\text{payoff} - V(S_0, 0; \sigma_i) = \int_0^T \Delta \, dS + \tfrac{1}{2} \int_0^T \Gamma_t S_t^2 \left(\sigma_t^2 - \sigma_i^2\right) dt.$$

Now take expectations under $\mathbb{Q}$. The hedging term is a martingale and drops out. The market price of the option is $\mathbb{E}^{\mathbb{Q}}[\text{payoff}]$, and by the definition of implied volatility that price is exactly $V(S_0, 0; \sigma_i)$. The left-hand side is therefore zero, and rearranging what is left:

$$\sigma_i^2 \;=\; \frac{\mathbb{E}^{\mathbb{Q}}\left[\displaystyle\int_0^T \Gamma_t S_t^2 \, \sigma_t^2 \, dt\right]}{\mathbb{E}^{\mathbb{Q}}\left[\displaystyle\int_0^T \Gamma_t S_t^2 \, dt\right]}, \qquad \Gamma_t = \Gamma^{\mathsf{BS}}(S_t, t; \sigma_i).$$

An option's implied variance is a weighted average of the variance the market prices in, with weights equal to the option's own dollar gamma along the paths the underlying might take. The relation is exact, not an approximation, given continuous paths and the usual integrability. It is implicit — the weights are computed at σ_i itself — but that does not weaken the reading, and the fixed point is unique in practice. The representation is standard: it underlies the delta-hedging P&L analysis of Carr and Madan (1998), the local-volatility averaging results in Gatheral (2006), and the survey of probabilistic interpretations in Lee (2005).

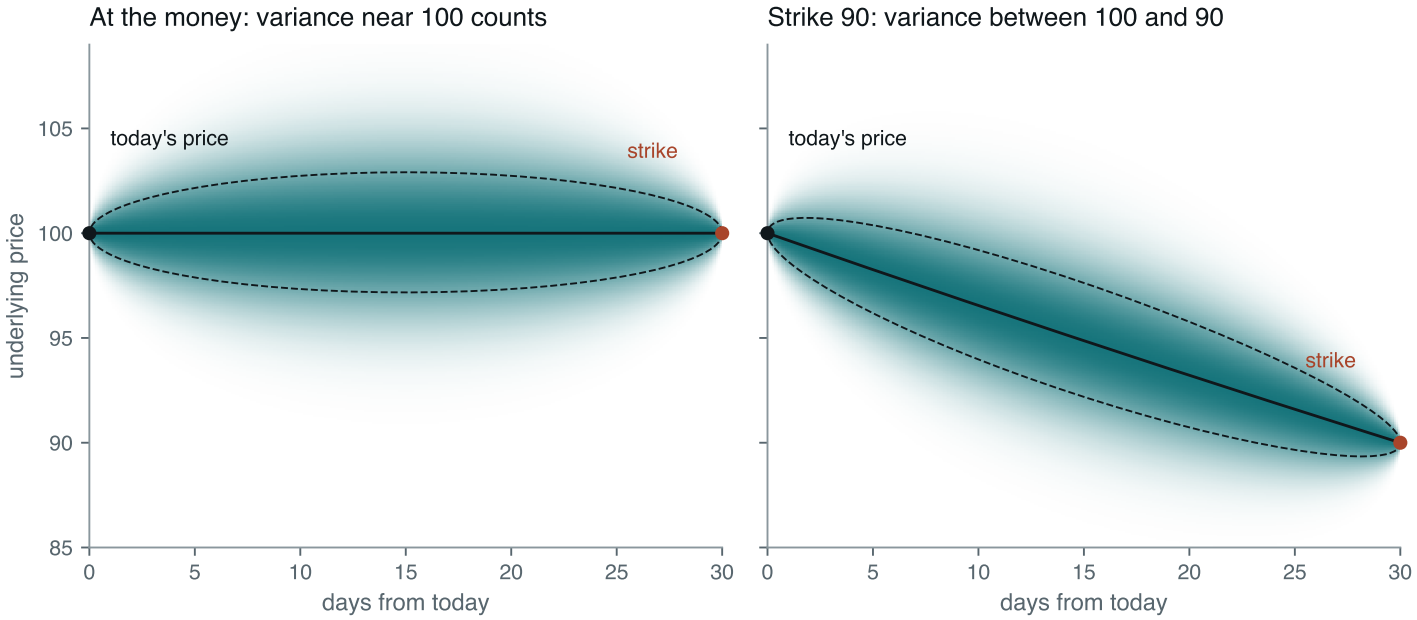
Four special cases show how much the equation explains on its own.

- **Constant volatility.** If $\sigma_t \equiv \sigma$, then $\sigma_i = \sigma$ for every strike and expiry: a flat surface. This is the Black–Scholes world, and it is the only world in which "the implied volatility" is a well-posed phrase.

- **Volatility that varies with time but not randomly.** If $\sigma_t = \sigma(t)$ is a known function of time, the weights cannot favour one level of the price over another, and the equation collapses to $\sigma_i^2 = \frac{1}{T}\int_0^T \sigma^2(t)\,dt$ — the root-mean-square of forward volatility. Every strike still has the same implied volatility: **a term structure with no smile**. This is the engine of §5.4.
- **Volatility that depends on the price level.** Now different strikes have their weights in different places (§4.4), so they average different volatilities and therefore differ: **a smile**. §8.2 makes this precise.
- **Volatility that is random but independent of the price.** The option price becomes an average of Black–Scholes prices over the distribution of average variance. Because the Black–Scholes price is convex in volatility away from the money (the second derivative, "vomma", is positive away from the money and negligible in a narrow band around the forward), averaging raises out-of-the-money prices more than at-the-money ones: **a symmetric smile**, with curvature growing in the uncertainty about volatility. Correlation between the price and its volatility tilts that smile into a skew (§8.4).

4.4 Where an option's weights live

The weights in the master equation are not abstract. Under Black–Scholes dynamics at σ_i, the expected dollar gamma at each time and price level has an exact and memorable form: it is a **Brownian bridge** from today's price to the strike on expiry day.



The picture is worth dwelling on, because it answers a question that trips up almost everyone: *which* volatility does an option's implied volatility refer to?

- For an **at-the-money** option, the weights form a lens centred on today's price. The option is a claim on variance delivered near the current price.
- For a **90-strike** option with the price at 100, the lens tilts: it runs from 100 today to 90 on expiry day, and at the midpoint of the option's life its one-standard-deviation band covers roughly 92.2 to 97.6. [Simulated] The option is a claim on the variance that occurs *along the way down*. If volatility in that region is higher than volatility near 100 — which is exactly what the leverage effect and crash risk imply for equities — then this option's implied volatility is higher. **That is a skew, arrived at without any model.**
- In **time**, the weights are flat: each third of the option's life contributes exactly a third of the total weight. [Simulated] This is initially surprising, since gamma explodes near expiry. The resolution is that gamma explodes only *near the strike*, and the probability of being near the strike late in the option's life shrinks at exactly the compensating rate.

The last point needs care, because it is easy to state a contradiction. *Unconditionally* the weight per unit time is flat. *Along a given path*, it is anything but: if the price happens to sit at the strike with two days left, the dollar gamma there is enormous and the variance delivered in those two days dominates the option's P&L. This is why a hedged option's outcome is path-dependent even when total realised volatility comes in exactly at implied (Dealer Hedging, section 4.6).

4.5 What the master equation explains

Almost everything in Part II falls out of the same three knobs — the weights, the measure, and the slice of time.

1. **Why each strike has its own number.** Different strikes weight different regions of price space. If volatility depends on the price level, their averages differ. The smile is not evidence that the market is confused about volatility; it is evidence that volatility is not a single number (§7, §8).
2. **Why "implied versus realised" is subtler than it looks.** The right comparison for a single option is implied variance against *dollar-gamma-weighted* realised variance, not against a plain close-to-close estimate over the same window (§11.8). Two paths with identical realised volatility hand a hedger very different P&L.
3. **Why flattening the weights gives a cleaner instrument.** If a portfolio's dollar gamma is the same at every price, the weighted average becomes a plain average, and the instrument pays realised variance with no dependence on where the price went. That portfolio exists, it is a $1/K^2$ strip of options, and it is a variance swap (§5.1).
4. **Why the term structure is an average of forward variances.** Slicing the integral in time gives additive total variance and the forward-variance reading (§5.4), and dropping a single dated lump of variance into the integral gives the implied move of an earnings announcement (§5.5).
5. **Why implied volatility is not a forecast.** Everything above is under $\mathbb{Q}$. Moving to the real-world measure introduces the variance risk premium, which is the entire subject of §6.
6. **Why the number is model-dependent in one specific way.** The weights come from Black–Scholes gamma at σ_i. That is a modelling choice baked into the definition, and it is the reason the same market can be described equally well by a local-volatility function, a stochastic-volatility model, or a jump model with very different behaviour (§8.4).

§4 Key takeaways

1. A delta-hedged option's P&L is half its dollar gamma times realised squared return minus the variance implied volatility charged for. Implied volatility is a rate, not an opinion.
2. The daily break-even move is $\sigma_i\sqrt{\Delta t}$ — 1.26% a trading day at 20% volatility. Losses beyond it grow quadratically.
3. Integrating the identity gives the master equation: implied variance is a dollar-gamma-weighted average of the variance priced in, exactly, for any continuous process.
4. Constant volatility gives a flat surface; deterministic time-varying volatility gives a term structure with no smile; price-dependent volatility gives a skew; random volatility independent of price gives a symmetric smile.
5. The weights are a Brownian bridge from today's price to the strike: an out-of-the-money option prices the variance along the path to its strike, which is where skew comes from.
6. Unconditionally the weights are flat in time; along any single path they are concentrated wherever the price sits near the strike late in the option's life.
7. Flattening the weights, changing the measure, or slicing time generates the variance swap, the forecast reading, and forward volatility respectively.

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5. The price of variance: variance swaps, the VIX and the term structure

5.1 Flattening the weights: the log contract and the variance swap

A single option's implied volatility mixes two things a volatility trader would rather keep apart: how much variance the market prices, and *where* that variance has to arrive to be worth anything (§4.4). The fix is to build a portfolio whose dollar gamma is the same at every price level. Then the weights in the master equation are flat, the weighted average becomes a plain average, and the position pays realised variance regardless of the path.

Such a portfolio exists, and its construction is the one piece of financial engineering every user of the VIX should be able to reproduce. A single option's dollar gamma is a bump centred on its strike, and the size of that bump scales with the square of the strike. So hold $1/K^2$ **of an option at every strike**: the bumps then have equal area, and they sum to a constant. Concretely, for options priced at one volatility,

$$\int_0^\infty \frac{1}{K^2}\,\Gamma_K(S)\,S^2\,dK \;=\; 1 \qquad \text{for every } S \text{ and every } \tau,$$

where Γ_K is the gamma of the option struck at K (this subscript is a strike, not a time). It follows in two lines from $\Gamma_K S^2 = S\varphi(d_1)/(\sigma\sqrt{\tau})$, the identity $S\varphi(d_1) = K\varphi(d_2)$, and changing the variable of integration from K to d_1.

Feed that portfolio into the hedged-P&L identity of §4.1. With dollar gamma constant at $2/T$ (the $1/K^2$ strip scaled by $2/T$), the P&L of the delta-hedged strip over the option's life is

$$\Pi = \frac{1}{T}\int_0^T \sigma_t^2\,dt \;-\; \sigma_{\text{VS}}^2,$$

realised variance minus a number fixed today. That is a **variance swap**: a contract that pays the difference between realised variance and a strike, with no dependence on the path beyond the variance itself. Its fair strike is the cost of the strip:

$$\sigma_{\text{VS}}^2 \;=\; \frac{2}{T}\int_0^\infty \frac{\text{OTM}(K)}{K^2}\,dK$$

(with zero rates; $\text{OTM}(K)$ is the out-of-the-money option at strike K — a put below the forward, a call above). Equivalently, the strip replicates a **log contract** paying $-\frac{2}{T}\ln(S_T/F)$, whose dollar gamma is constant by inspection (Neuberger, 1994); the general statement that any payoff can be decomposed into a bond, a forward and a strip of options with weight $f''(K)$ is the spanning formula of Carr and Madan (2001). The practitioner reference that made this standard is Demeterfi, Derman, Kamal and Zou (1999), and the theoretical statement — that this strip gives the $\mathbb{Q}$-expected integrated variance for *any* continuous process, with no model at all — is Britten-Jones and Neuberger (2000).

Two intuitions are worth carrying away. **Low strikes get more contracts**: $1/K^2$ weighting means far more puts than calls by contract count, which is why a variance-swap rate is so sensitive to the price of deep out-of-the-money puts. And **the strip is long convexity everywhere**, so it is the natural instrument for anyone who wants to own or sell "volatility" without also taking a view on where the price will be.

5.2 Model-free implied variance, and the VIX

Because the replication is model-free, the strip's cost is a direct reading of the market's priced variance. Computing it continuously and publishing the answer is exactly what a volatility index does. Cboe's VIX methodology sums out-of-the-money SPX option mid-quotes as

$$\sigma^2 = \frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2}\,e^{RT}\,Q(K_i) \;-\; \frac{1}{T}\left(\frac{F}{K_0}-1\right)^2,$$

the discretised version of the integral above, in Cboe's own notation: R is the constant-maturity risk-free rate to that expiry (the r of the notation block, sourced as below), and $Q(K_i)$ is the midpoint quote at strike K_i — an option price, not to be confused with the measure $\mathbb{Q}$. The remaining term corrects for the fact that the sum starts at K_0, the first listed strike below the forward, rather than exactly at the forward. The published index then interpolates the two expiries that bracket 30 days — linearly in **total variance**, weighted by minutes — and annualises: $\text{VIX} = 100\sqrt{(\text{30-day interpolated variance})}$.

The implementation details are worth knowing, because every one of them is a place your own calculation can diverge from the published number:

Rule	What Cboe does
Universe	Monthly AM-settled SPX options and weekly PM-settled SPXW options, quotes from Cboe's own exchange
Which options	Out-of-the-money only: puts below K_0, calls above, and the average of the put and call at K_0
Price used	The midpoint of the bid–ask spread; only options with a non-zero bid
Strip truncation	Stop after **two consecutive strikes with zero bids**, and ignore anything beyond
Forward	Implied from put–call parity at the strike where the call and put prices are closest
Rates	Constant-maturity Treasury yields, cubic-spline interpolated to each expiry
Term interpolation	Linear in total variance, by minutes, to a constant 30 days

The index has a history worth knowing too. The original 1993 VIX (now published as VXO) was an average of Black–Scholes implied volatilities of near-the-money S&P 100 options (Whaley, 1993), which is a different object: Carr and Wu (2006) show that the old index approximates a *volatility*-swap rate while the 2003 redefinition targets a *variance*-swap rate. The distinction is not academic; it is most of §5.3.

5.3 Why the VIX is not at-the-money implied volatility

A VIX-style number and an at-the-money implied volatility are routinely treated as interchangeable. They are not, and the gap between them has two separate causes.

Cause 1: the whole smile enters, not just the middle. The variance-swap rate is an average of the smile's implied variances, weighted by the standard normal density in the standardised strike variable $z = -d_2$:

$$\sigma^2_{\text{VS}} \;=\; \int_{-\infty}^{\infty} \varphi(z)\, \sigma_i^2\big(k(z)\big)\, dz.$$

This identity — due to Gatheral (2006), and exact when paths are continuous and the map from strike to z is monotone — is the cleanest statement of the relationship between "the VIX" and "the smile". I checked it numerically against the $1/K^2$ strip on three stylised 30-day smiles, all with a 20.0% at-the-money volatility: [Simulated]

Smile	At-the-money vol	Variance-swap vol from the strip	From the normal-weighted smile
Flat	20.00%	20.00%	20.00%
Symmetric smile	20.00%	21.98%	21.98%
Equity-index skew	19.99%	23.55%	23.55%

The two computations agree to three decimals, and the message is plain: **wings raise a variance-swap rate above at-the-money volatility, and a steep skew raises it a lot.** Three and a half volatility points of the gap in the last row is not a forecast of anything; it is the price of the out-of-the-money puts.

Cause 2: a square root of an average is not an average of square roots. A variance swap pays variance; a volatility swap pays volatility. Since the square root is concave,

$$\mathbb{E}\big[\sqrt{\text{variance}}\big] \;\le\; \sqrt{\mathbb{E}[\text{variance}]},$$

so the fair volatility strike sits *below* the square root of the fair variance strike by roughly $\text{Var}\big(\text{realised variance}\big)/\big(8\,\sigma^3_{\text{VS}}\big)$. With a 20% variance-swap rate and a standard deviation of realised variance equal to half its mean, that convexity adjustment is about 0.6 volatility points. At-the-money implied volatility is much closer to the volatility-swap rate than to the variance-swap rate (Carr and Lee, 2009), which is the formal version of the rough practitioner statement that the VIX "sits above" 30-day at-the-money volatility. [Practice]

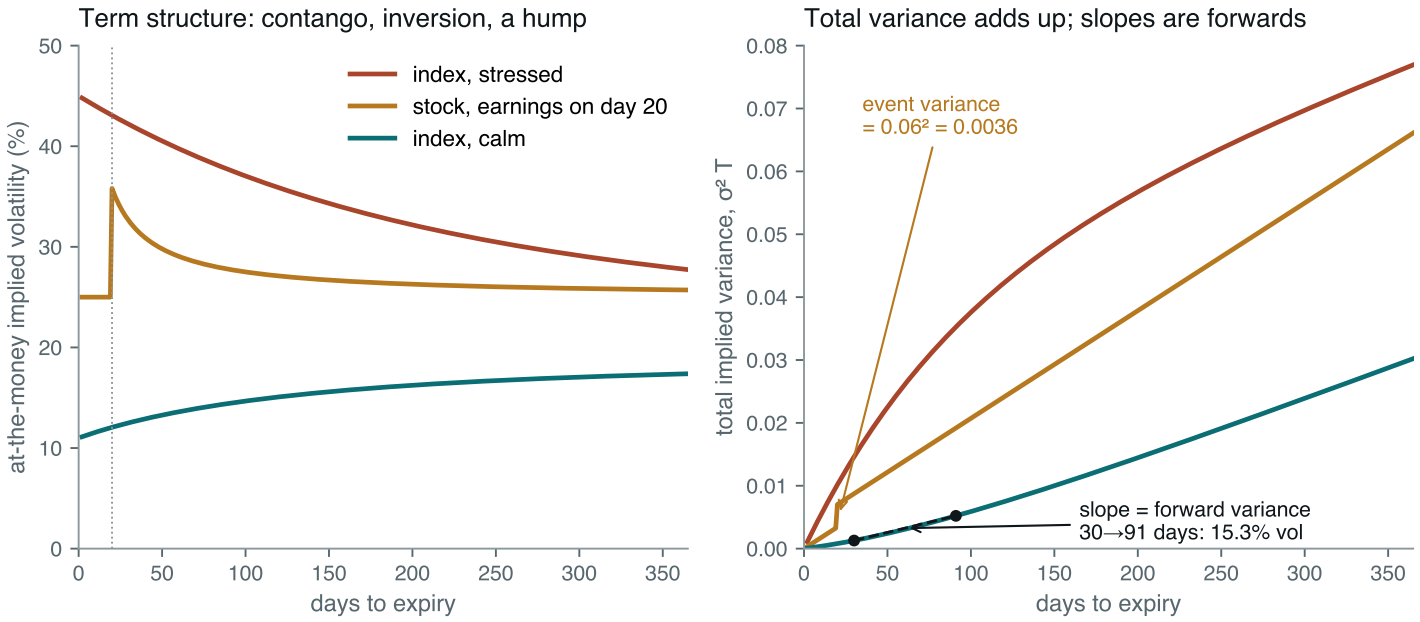
The practical consequences are immediate. Do not compare a VIX-style index on one underlying with an at-the-money volatility on another. Do not conclude that "volatility rose" when a widening skew lifted a variance index while at-the-money volatility was flat. And when you need a clean level, say which object you mean.

5.4 Total variance adds up: the term structure

Set the strike aside and slice the master equation in time instead. Because variance accumulates, total implied variance is additive: the variance priced between two dates is the difference of the total variances, and the **forward variance** between expiries T_1 and T_2 is

$$\sigma^2_{\text{fwd}}(T_1, T_2) \;=\; \frac{w(T_2) - w(T_1)}{T_2 - T_1} \;=\; \frac{\sigma_i^2(T_2)\,T_2 - \sigma_i^2(T_1)\,T_1}{T_2 - T_1}.$$

One month at 20% and two months at 22% imply a forward volatility of 23.8% for the month in between — a number several points above either quoted volatility, which is the usual source of surprise the first time someone computes one.



The left panel shows the three shapes that account for most of what term structures ever do. [Simulated]

- **Contango.** In calm markets the curve slopes up: 11.4% at a week, 12.5% at a month, 17.4% at a year in the stylised example. Short-dated variance is low, and the market prices mean reversion back to a higher long-run level.
- **Backwardation.** In stressed markets it inverts: 44.3% at a week falling to 27.7% at a year. Today is frightening; the market does not expect it to last.
- **An event hump.** A stock with an earnings announcement on day 20 has a flat 25% term structure for expiries *before* the event and a jump to 35.4% for the first expiry *after* it, decaying slowly thereafter as the fixed event variance is spread over more days.

The right panel is the same information in the coordinate that makes it obvious. In total variance, each curve is increasing, its slope at any point is the forward variance, and the event is a vertical step of exactly the event's variance. This is also where the no-arbitrage condition lives: **total variance must be non-decreasing in expiry** at a fixed moneyness, because forward variance cannot be negative. A one-month option at 30% ($w = 0.0075$) alongside a two-month option at 20% ($w = 0.0067$) is not a steep term structure, it is an arbitrage.

5.5 Events: extracting the implied move

The event hump is not a curiosity; it is how the options market prices scheduled information, and reading it backwards is a standard desk calculation. Model total variance as a diffusive part plus a lump that arrives on the event date:

$$w(T) = \sigma^2_{\text{base}}\, T + \sigma^2_{\text{event}} \cdot \mathbf{1}\{T \geq t_{\text{event}}\}.$$

With expiries either side of the event, the lump is a difference: $\sigma^2_{\text{event}} = w(T_2) - w(T_1) - \sigma^2_{\text{base}}(T_2 - T_1)$. In the stylised example, a 14-day option at 25.0% and a 28-day option at 33.1% recover an event standard deviation of exactly 6.00%, and hence an expected absolute move of $6.00\% \times \sqrt{2/\pi} = 4.79\%$. [Simulated] With only one post-event expiry and an assumed base volatility, the same arithmetic works with more assumption in it: a 30-day option at 40% against a 25% base implies an event standard deviation of 8.95% and an expected absolute move of 7.1%.

This is the number quoted as "the options market is pricing a 7% move on earnings", and §13.3 covers how it is used and misused. Two things about it are worth flagging now. It is a **risk-neutral** expectation, so it carries whatever premium sellers of event risk demand — and there is evidence that earnings-event volatility is priced at a premium to what is realised, along with evidence that the implied event move is genuinely informative about the size of the move that follows (Dubinsky, Johannes, Kaeck and Seeger, 2019). And the "vol crush" after the announcement is not sentiment: the variance has been spent, the lump leaves the remaining total variance, and implied volatility mechanically drops back to the base level.

The same decomposition explains why desks run a **variance clock** rather than a calendar: weekends and holidays carry little variance, event days carry a lot, and pricing a Friday-to-Monday option on calendar days overstates the variance the market expects. [Practice]

5.6 Where the replication breaks

The variance-swap replication is exact only under assumptions, and every one of them fails somewhere.

- **Jumps.** The strip replicates the log contract, and the log contract tracks quadratic variation only for continuous paths. A single jump of $\Delta \ln S = -20\%$ (a drop in price of about 18%) contributes $(\Delta \ln S)^2 = 0.04$ to realised variance but only 0.0375 to the log contract — the strip captures 94% of it, the error being third order in the jump size. Jump risk therefore drives a small wedge between a variance-swap rate and a VIX-style index, discussed in Carr and Wu (2009) and Carr and Lee (2009).
- **Truncation.** The integral runs over all strikes; the market lists finitely many, and the VIX rule stops after two consecutive zero bids. Truncation always *understates* variance, and it understates it most when the tails are expensive — that is, in stress. Andersen, Bondarenko and Gonzalez-Perez (2015) show that the strike range entering the index varies over time and injects noise of its own.
- **Discretisation.** Strikes are spaced, so the sum approximates the integral. Wider spacing biases the answer, which matters more for single stocks than for SPX.
- **Quotes, not trades.** The index is built from bid–ask midpoints. Anything that widens quotes lifts the index even if nothing trades — the mechanism behind the August 2024 VIX spike (Todorov and Vilkov, 2024).
- **Settlement.** VIX derivatives settle against a special opening auction in the option strip, whose mechanics have attracted a [Contested] literature on manipulation (Griffin and Shams, 2018; Cboe disputes the interpretation).
- **Discrete monitoring.** Contractual realised variance is computed from daily closing log returns, not from the continuous quadratic variation the replication assumes.

None of this makes the construction unusable — it is the best-understood object in the volatility world — but it does mean that a VIX-style number computed from your own data and the published index will differ in the third decimal for boring reasons, and in the first decimal in a crisis for interesting ones.

§5 Key takeaways

1. A strip of options weighted $1/K^2$ has constant dollar gamma, so hedging it pays realised variance against a fixed strike: that is a variance swap.
2. Its fair strike is model-free — it holds for any continuous process — which is why a volatility index can be published without a pricing model.
3. The VIX is that strip, discretised: out-of-the-money mid-quotes, two expiries bracketing 30 days, interpolated linearly in total variance.
4. A variance-swap rate exceeds at-the-money volatility for two reasons: the wings enter the average (3.5 points on a stylised index skew), and variance swaps pay variance rather than volatility (about 0.6 points of convexity).
5. Total implied variance is additive in time; its slope is forward variance, and total variance must not decrease with expiry.
6. A scheduled event is a lump of variance in the term structure; differencing two expiries recovers the market's expected move, and the post-event "vol crush" is that lump being spent.
7. Jumps, truncation, discrete strikes and quote-based pricing all put a wedge between the theory and a published index.

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6. A forecast with a premium attached

6.1 Two probability measures

Everything in §4 and §5 was computed under $\mathbb{Q}$, the pricing measure. The distinction between $\mathbb{Q}$ and the real-world measure $\mathbb{P}$ is the single most important thing to hold on to when reading implied volatility as a forecast, and it has a one-line intuition.

Prices are expectations of payoffs *weighted by how much a dollar is worth in each state of the world*. A dollar delivered in a crash is worth more than a dollar delivered in a rally, because it arrives when the holder is poor. Rolling that weighting into the probabilities produces $\mathbb{Q}$: a set of probabilities that overweights bad states relative to how often they actually happen. Implied volatility, being a price, inherits that overweighting. So

$$\underbrace{\sigma_i^2}_{\text{what options price}} = \underbrace{\mathbb{E}^{\mathbb{P}}[\text{variance}]}_{\text{what you expect to happen}} + \underbrace{\text{compensation for bearing the risk}}_{\text{§6.4, mechanisms 1 and 3}} + \underbrace{\text{demand pressure and frictions}}_{\text{§6.4, mechanism 2}}.$$

Two conventions before the evidence. The last two terms are not separately identifiable from prices alone — both just make options expensive relative to expected variance — so this document follows the literature and bundles them: the **variance risk premium** is defined as implied minus expected realised variance, the whole gap, so a positive premium means options are expensive relative to what happens; much of the literature, including Carr and Wu (2009), defines it with the opposite sign as realised minus implied. And premia should be compared in **variance** units; the volatility-point difference that practitioners quote mixes in the convexity of §5.3.

6.2 How good a forecast is implied volatility?

The empirical literature on this question is unusually clean, because it produced a famous disagreement that was subsequently resolved.

- Latané and Rendleman (1976) opened the question within three years of the Black–Scholes paper, finding that implied standard deviations were related to subsequent realised volatility.
- Canina and Figlewski (1993) then found the opposite, and emphatically: for S&P 100 index options, implied volatility had "virtually no correlation" with subsequent realised volatility and did not even subsume information in recent historical volatility.
- Christensen and Prabhala (1998) revisited the same market over a longer sample with non-overlapping monthly observations, instrumenting implied volatility to deal with measurement error, and found the reverse: implied volatility outperforms past realised volatility and subsumes it in some specifications. They attribute the earlier result partly to a regime shift around the 1987 crash and partly to the statistical treatment.
- Jorion (1995) reached a similar conclusion in currency options: implied volatility beats time-series models, *and* is a biased forecast.
- Jiang and Tian (2005) showed that the model-free (variance-swap) measure of §5 subsumes both Black–Scholes implied volatility and past realised volatility.
- Poon and Granger (2003), reviewing 93 studies, found option-implied forecasts generally the most accurate class of forecast among those compared.

[Fact] Implied volatility is an informative but biased forecast of subsequent realised volatility: informative in that it beats or subsumes history-based forecasts at horizons of a month or so, biased in that it sits above what is realised on average.

Two methodological points matter if you intend to test this yourself. Realised variance is a noisy proxy for the thing being forecast, so forecast comparisons need a loss function that is robust to that noise (Patton, 2011). And overlapping windows — regressing 30-day realised volatility on daily observations of 30-day implied volatility — induce serial correlation that makes standard errors far too small.

6.3 The variance risk premium

The bias has a name, a sign, a size and a long list of citations.

[Fact] For equity indices, implied variance exceeds subsequently realised variance on average, by a wide margin, and the gap is one of the more robust facts in asset pricing. Over 1990–2024 the VIX averaged 19.6% against about 15.5% for the S&P 500's subsequent 30-day realised volatility — roughly four volatility points (Horstmeyer, Handley and Manuel, 2024, a practitioner piece whose methodology is not fully described; treat the exact figure as indicative and the sign as settled). The academic measurements agree: Carr and Wu (2009) find strongly negative variance risk premia on index variance swaps, Bakshi and Kapadia (2003) find that buyers of delta-hedged index options lose on average, and Coval and Shumway (2001) find that zero-beta at-the-money index straddles lose roughly 3% a week.

Four qualifications keep this from being a licence to print money.

- **The premium is concentrated in indices.** Single-stock variance risk premia are much smaller and often statistically indistinguishable from zero (Carr and Wu, 2009). Driessen, Maenhout and Vilkov (2009) explain the difference: what is expensive about index options is correlation, not volatility.
- **It is concentrated at the short end.** Dew-Becker, Giglio, Le and Rodriguez (2017) find that between 1996 and 2014 it was costless on average to hedge *news about future variance* at horizons from one quarter to fourteen years: only unexpected, transitory realised variance was priced.
- **It varies enormously over time, and that variation is informative.** Bollerslev, Tauchen and Zhou (2009) show the premium predicts aggregate stock returns, most strongly at a quarterly horizon. Bekaert and Hoerova (2014) decompose

the squared VIX into a conditional-variance part and a premium part and find that the two components carry different predictive content for returns, economic activity and financial instability.

- **The average hides the shape.** Selling variance earns a small positive amount most months and loses catastrophically in a few. The premium is the fee for holding that distribution, not an anomaly, and the February 2018 collapse of short-volatility products is what its left tail looks like (Dealer Hedging, section 8.4).

6.4 Why the premium exists

Three mechanisms, none of which excludes the others.

1. **Volatility is insurance.** A long-volatility position pays off precisely when markets fall and portfolios hurt, so it has a strongly negative beta to bad states. Investors will accept a negative expected return on such an asset for the same reason they pay for insurance. [Fact] for the hedging property; the magnitude it justifies is [Contested].
2. **Demand pressure meets constrained intermediaries.** End users are net buyers of index puts and net sellers of single-stock options; dealers take the other side and cannot hedge the residual risk perfectly (jumps, discrete rehedging, liquidity), so they charge for the inventory they carry (Bollen and Whaley, 2004; Gârleanu, Pedersen and Poteshman, 2009). This also explains why the premium is largest exactly where demand is largest — index downside strikes.
3. **Correlation risk.** For an index specifically, part of the premium is payment for the risk that correlations rise in a sell-off, which is what makes index variance spike more than component variance (Driessen, Maenhout and Vilkov, 2009).

Why it is not arbitraged away is a separate question with a practical answer: the trade that harvests it has small steady gains and rare enormous losses, requires capital against those losses, and is hardest to hold precisely when it is most profitable to initiate.

6.5 Using a biased forecast

Given all of the above, here is what I would actually do with implied volatility when the goal is a forecast.

- **Debias before use.** Regress realised variance on implied variance (and on a history-based forecast such as HAR) and use the fitted value, rather than implied variance raw. The slope is typically below one and the intercept positive. [Practice]
- **Work in variance, convert at the end.** Premia, forecasts and regressions are linear in variance and not in volatility.
- **Match the horizon exactly.** A 30-day implied volatility forecasts the next 30 calendar days — roughly 21 trading days — and nothing else. Most "implied volatility doesn't work" results I have seen were horizon mismatches.
- **Do not trade the gap unless it exceeds the premium.** Implied volatility above your forecast is the normal state of the world. The tradable signal is the gap *relative to its own history*, net of hedging error and costs (§13.2).
- **For risk management, the bias is a feature.** An input that reacts within minutes to a shock and errs on the high side is what you want in a stress calculation, as long as you do not also call it an unbiased expectation (§13.4).

One deeper question is whether the entire real-world distribution can be recovered from option prices rather than just corrected on average. Ross (2015) showed that under strong assumptions on preferences and state dynamics it can; Borovička, Hansen and Scheinkman (2016) showed that those assumptions do most of the work and that the recovered measure generally is not the physical one. [Contested], and my read is that the negative case is the stronger one: treat option-implied distributions as prices, not probabilities.

§6 Key takeaways

1. Implied volatility is computed under the pricing measure, which overweights bad states. A forecast requires removing that weighting.
2. The early finding that implied volatility has no forecasting power was overturned: it beats or subsumes history-based forecasts, and the model-free version subsumes the Black–Scholes one.
3. It is nonetheless biased upward: the S&P 500's implied volatility has averaged about four points above subsequent realised volatility since 1990.
4. The premium is concentrated in indices rather than single stocks, and at short horizons rather than long ones.
5. It is compensation for a genuinely bad payoff distribution — small steady gains, rare enormous losses — not an anomaly.
6. Debias by regression, work in variance units, match horizons exactly, and do not mistake a normal premium for a trading signal.
7. Recovering real-world probabilities from option prices is contested; treat implied distributions as prices.

7. A distribution in lognormal units: the smile

7.1 A flat smile is a lognormal distribution

Fix an expiry and look across strikes. If every strike has the same implied volatility, then every option is priced by the same Black–Scholes formula with the same σ, and the distribution of the price at expiry that those prices imply is exactly lognormal with log standard deviation $\sigma\sqrt{T}$. A flat smile *is* a lognormal distribution, restated.

So every departure from flatness is a departure from lognormality, and the smile is a picture of that departure in the most convenient possible coordinates: the vertical axis says, strike by strike, *how wide a lognormal distribution would have to be to price this particular option correctly*. An out-of-the-money put at a high implied volatility is the market saying that the amount of probability mass below that strike (weighted by how far below) is more than a lognormal fitted to at-the-money options would put there.

7.2 Butterflies are probabilities: Breeden–Litzenberger

The connection between prices and the distribution is not an analogy; it is an identity, and the cleanest derivation of it is a trade.

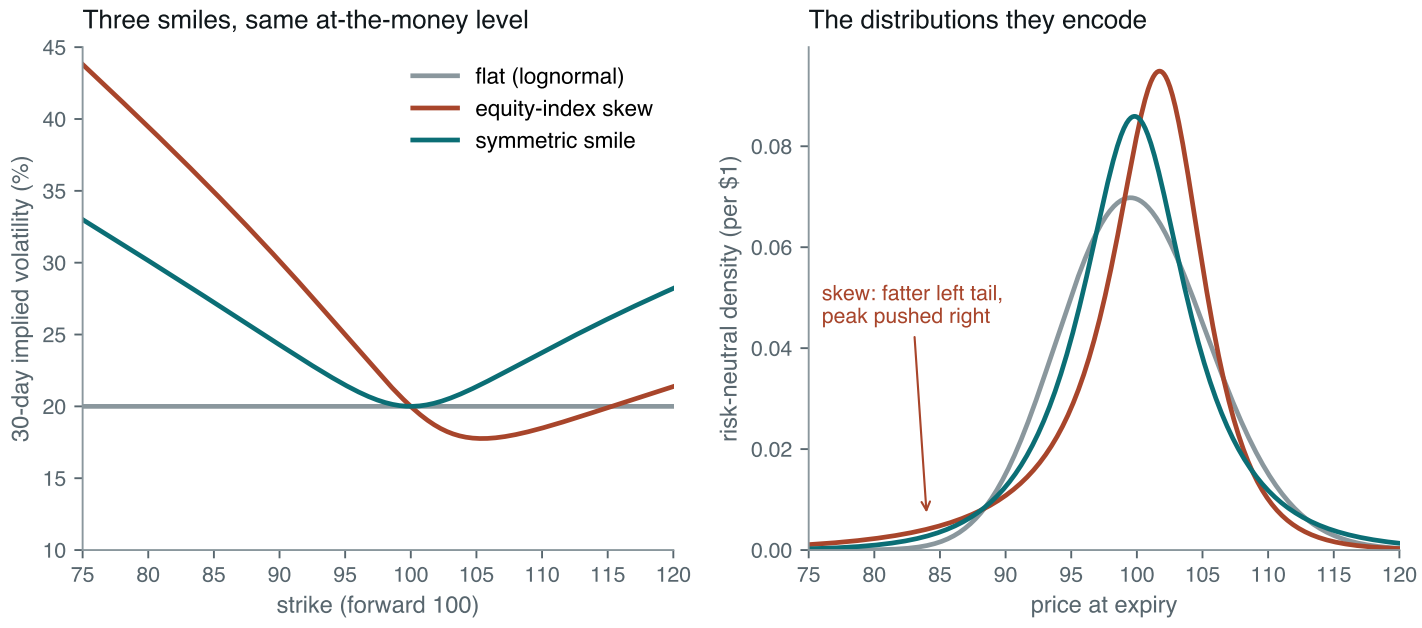
Buy one call at $K-h$, sell two at K, buy one at $K+h$. This **butterfly spread** pays nothing unless the price finishes between $K-h$ and $K+h$; inside that range it pays a triangle peaking at h when the price finishes exactly at K. The triangle's area is h^2. So for small h, the spread is a bet that pays about h^2 if the price lands near K, and its price is the discounted probability of landing there times h^2. Dividing by h^2 and taking the limit,

$$f_T(K) \;=\; \frac{1}{D}\,\frac{\partial^2 C}{\partial K^2},$$

the risk-neutral density of the price at expiry (Breeden and Litzenberger, 1978). The same logic one derivative earlier gives the digital: a call spread $\big(C(K)-C(K+h)\big)/h$ pays 1 if the price exceeds K, so

$$\mathbb{Q}(S_T > K) \;=\; -\frac{1}{D}\,\frac{\partial C}{\partial K}.$$

Option prices across strikes are the distribution. Not an estimate of it, not a model of it: a different encoding of the same information. The smile is that encoding converted into volatility units, and §7.4 is what happens when people forget that the conversion is nonlinear.



7.3 Reading the shape: level, slope and curvature

The figure shows three 30-day smiles constructed to have the *same* 20% at-the-money volatility, and the densities they encode. [Simulated] The numbers underneath are the translation table.

	Flat	Equity-index skew	Symmetric smile
Implied vol at strike 80	20.0%	39.5%	30.2%
Implied vol at strike 90	20.0%	30.1%	24.3%
Implied vol at strike 100	20.0%	20.0%	20.0%
Implied vol at strike 110	20.0%	18.5%	23.8%
Standard deviation of the log return	5.73%	6.90%	6.30%
Skewness of the log return	0.00	−2.00	−0.08
Excess kurtosis	0.00	+8.97	+2.66
Most likely price at expiry	99.5	101.7	99.8

Three readings, in order of how often they are used:

- **Level → width.** The at-the-money volatility sets the scale of the distribution.
- **Slope → skewness.** A smile that falls from left to right encodes a left-skewed distribution. The index skew here has skewness -2.0 against zero for the lognormal. Note the counter-intuitive consequence visible in the figure: a left-skewed distribution has its *mode to the right* of the lognormal's (101.7 versus 99.5). Mass is pulled out of the middle-left and pushed into both the far-left tail and the region just above the forward. The market is pricing "usually drifts up a little, occasionally collapses".
- **Curvature → tails.** A U-shaped smile encodes excess kurtosis: fat tails on both sides *and* a sharper peak, since the mass has to come from somewhere. The symmetric smile here has excess kurtosis of 2.7 with essentially no skewness.

To first order, and only for modest departures from lognormality, this is captured by a Gram–Charlier expansion of the density in the spirit of Corrado and Su (1996):

$$\sigma_i(d) \;\approx\; \bar{\sigma}\left[1 - \frac{\gamma_1}{6}\,d - \frac{\gamma_2}{24}\,(1-d^2)\right],$$

where $d \approx d_2$ is the standardised distance from the strike to the forward — like d_2, it *falls* as the strike rises — and γ_1 the skewness and γ_2 the excess kurtosis of the log return to expiry. Negative skewness tilts the smile down to the right; positive excess kurtosis lifts both wings and depresses the middle. Treat this as a qualitative map, not a calibration tool: with the skewness of -2 that a real index smile implies, the first-order expansion is badly out of its range of validity.

The model-free way to get the same information is to price the moments directly. Bakshi, Kapadia and Madan (2003) show that risk-neutral variance, skewness and kurtosis are each the price of a specific payoff, and therefore each replicable by a strip of options with its own weighting function — the variance contract of §5.1 being the first of the family. Cboe's SKEW index is a published transformation of the 30-day risk-neutral skewness computed this way (100 minus ten times the skewness), which is why it rises when out-of-the-money puts richen relative to calls.

7.4 Probabilities from the smile, and the skew correction

Here is the single most common quantitative mistake made with implied volatility.

An analyst wants the probability that the index finishes below 90. They look up the implied volatility at the 90 strike — 30.1% in the index-skew example — plug it into $N(-d_2)$, and report 11.96%. The correct answer from the same smile is **6.83%**. [Simulated] They have overstated the probability by three quarters.

The error is conceptual, not arithmetic. The probability is the *slope* of the price curve across strikes, not a function of the *level* of one point on it. Differentiating through the smile,

$$\mathbb{Q}(S_T > K) \;=\; N(d_2) \;-\; \frac{\nu}{D}\,\frac{\partial \sigma_i}{\partial K},$$

where ν is the call's vega and both terms are evaluated at the strike's own implied volatility. The second term is the skew correction, and with a downward-sloping smile it is positive: **the probability of finishing above any strike is higher than $N(d_2)$ suggests, and the probability of finishing below it is lower.** Plugging a high out-of-the-money put volatility into a Black–Scholes probability treats the entire distribution as that wide, when in fact only the tail is.

The full comparison on the stylised smiles: [Simulated]

	Flat (lognormal)	Equity-index skew	Symmetric smile
$\mathbb{Q}(S_T < 90)$, computed correctly	3.52%	6.83%	4.93%
Naive $N(-d_2)$ at that strike's own vol	3.52%	11.96%	6.96%
$\mathbb{Q}(S_T > 105)$, computed correctly	18.95%	16.65%	16.63%
Naive $N(d_2)$ at that strike's own vol	18.95%	16.28%	20.41%

The skew roughly doubles the true probability of a 10% fall relative to a lognormal fitted at the money (3.5% to 6.8%) — and the naive calculation nearly doubles it again. Both errors are large, and they point in opposite directions, which is how a desk can have two analysts disagreeing by a factor of three about the same strike.

Two further warnings attach to any probability read off options. It is a $\mathbb{Q}$-probability, inflated relative to reality by the risk premium of §6, so it is an upper bound on the real-world chance of the bad outcome rather than an estimate of it. And delta is not a probability either: $\Delta = N(d_1)$, not $N(d_2)$, and both differ from the skew-corrected number above. The "probability of profit" displayed by retail options platforms is usually one of these three quantities, rarely labelled. [Practice]

7.5 How steep the wings can be

There is a hard limit on how fast a smile can rise in the tails, and it is a no-arbitrage result rather than a modelling convention. Lee (2004) showed that total implied variance can grow at most *linearly* in log-moneyness:

$$\limsup_{k \to \pm\infty} \frac{w(k)}{|k|} \;=\; \beta_{R,L} \in [0, 2],$$

and — the elegant part — the slope is pinned to the existence of moments. If the risk-neutral distribution has finite moments of every order, the wings are asymptotically flat in total-variance terms; the maximal slope of 2 corresponds to a distribution whose moments blow up immediately. In volatility terms the bound says $\sigma_i(k) \lesssim \sqrt{2|k|/T}$: at one month, no strike at half the forward can have an implied volatility above about 411%; at one year, above about 118%.

The bound looks generous, and for quoted strikes it usually is. It bites in two places that matter. First, **any smile parameterisation that is quadratic in log-moneyness violates it eventually**, which is why fitted polynomials produce arbitrage when extrapolated and why SVI's wings are linear by construction (§10.4). Second, every computation that integrates across all strikes — variance-swap rates, densities, risk-neutral moments — depends on how the wings are extrapolated beyond the last quoted strike, and a Lee-consistent extrapolation is the minimum standard.

7.6 Why smiles look the way they do, by market

The shape of a smile is a fact about the market's participants and the underlying's dynamics, and it differs systematically by asset class.

Market	Typical shape	Main mechanisms	Status
Equity indices	Steep negative skew, mild right wing	Leverage and volatility feedback; crash-risk pricing; net end-user demand for puts against constrained dealers; correlation rising in sell-offs	[Fact] for the shape, [Contested] for the split
Single stocks	Flatter skew, more symmetric, event humps	Idiosyncratic upside jumps (takeovers, results) offset downside skew; less put demand	[Fact]
Foreign exchange	Broadly symmetric smile with a tilt that follows crash risk	Carry currencies price crash risk on the high-yield side	[Fact]
Commodities	Often positive skew (calls richer)	Supply shocks spike prices upward; storage constraints	[Practice]
Rates	Depends on the ruler; normal-volatility quoting flattens it	Zero-bound effects; lognormal quoting manufactures skew (§2.5)	[Practice]

For equity indices specifically, the persistent downward skew dates from the 1987 crash (Rubinstein, 1994) and four mechanisms compete to explain it:

1. **Leverage.** A fall in equity raises a firm's debt-to-equity ratio and therefore the volatility of its equity — the explanation Fischer Black proposed in 1976, and measured by Christie (1982). Real, but too small in magnitude to explain the observed asymmetry.
2. **Volatility feedback.** If volatility rises, required returns rise, so prices fall immediately: causation running the other way (Campbell and Hentschel, 1992).
3. **Crash risk.** The distribution genuinely has a fat left tail, and models with jumps fitted to option prices say the market prices a meaningful chance of a large drop (Bates, 2000).
4. **Demand and intermediation.** End users buy index puts for protection; dealers who absorb that demand cannot hedge it perfectly and charge for the risk (Bollen and Whaley, 2004; Gârleanu, Pedersen and Poteshman, 2009).

My own read, stated as a view rather than a finding: the index skew is mostly the price of crash insurance sold by constrained intermediaries, sitting on top of a real but smaller asymmetry in the physical distribution. The strongest evidence for the demand channel is that skew is steepest exactly where end-user demand concentrates, and that single-stock skews — where end users are net *sellers* of options — are much flatter (Bakshi, Kapadia and Madan, 2003). For foreign exchange, the analogous evidence is that risk reversals on carry-trade currencies price the crash risk that those trades actually exhibit (Brunnermeier, Nagel and Pedersen, 2008).

§7 Key takeaways

1. A flat smile is exactly a lognormal distribution; every departure from flat is a departure from lognormality.
2. The second derivative of call prices across strikes is the risk-neutral density, and the first derivative is the digital probability. Option prices *are* the distribution.
3. Level maps to width, slope to skewness, curvature to fat tails. A left-skewed distribution has a mode above the lognormal's, not below it.
4. Probabilities are slopes, not levels: plugging a strike's own implied volatility into $N(-d_2)$ overstated the chance of a 10% fall by three quarters in the stylised example. The correct reading adds a skew term.
5. Any probability from options is a $\mathbb{Q}$-probability, and therefore an upper bound on the real-world chance of the bad state, not an estimate of it.
6. No-arbitrage caps wing growth at slope 2 in total variance, which kills quadratic smile fits and disciplines every extrapolation.

7. Index skew is steep and persistent post-1987; single stocks are flatter, commodities often tilt the other way, and part of any "skew" is an artefact of the quoting ruler.

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8. Local volatility, and how the surface moves

8.1 Local volatility: the one diffusion that fits

The master equation of §4.3 says an option's implied variance averages the variance priced along its own paths. That invites an inverse question: **is there a single process whose options reproduce the entire observed surface?**

Dupire (1994) and Derman and Kani (1994) answered yes, and uniquely, within the class of diffusions whose volatility is a deterministic function of price and time. That function is **local volatility** $\sigma_{\text{loc}}(S,t)$, and it is recovered from call prices by

$$\sigma^2_{\text{loc}}(K,T) \;=\; \frac{\partial C/\partial T}{\tfrac{1}{2}K^2\,\partial^2 C/\partial K^2}$$

(with zero rates and dividends). Read it as a ratio of two things §5 and §7 already named. The numerator is how much option value one extra instant of expiry adds: the variance the market prices *at that strike and date*. The denominator is the risk-neutral density at that strike, times $K^2/2$. So

local variance = the value of one more instant of time, per unit of probability of being there.

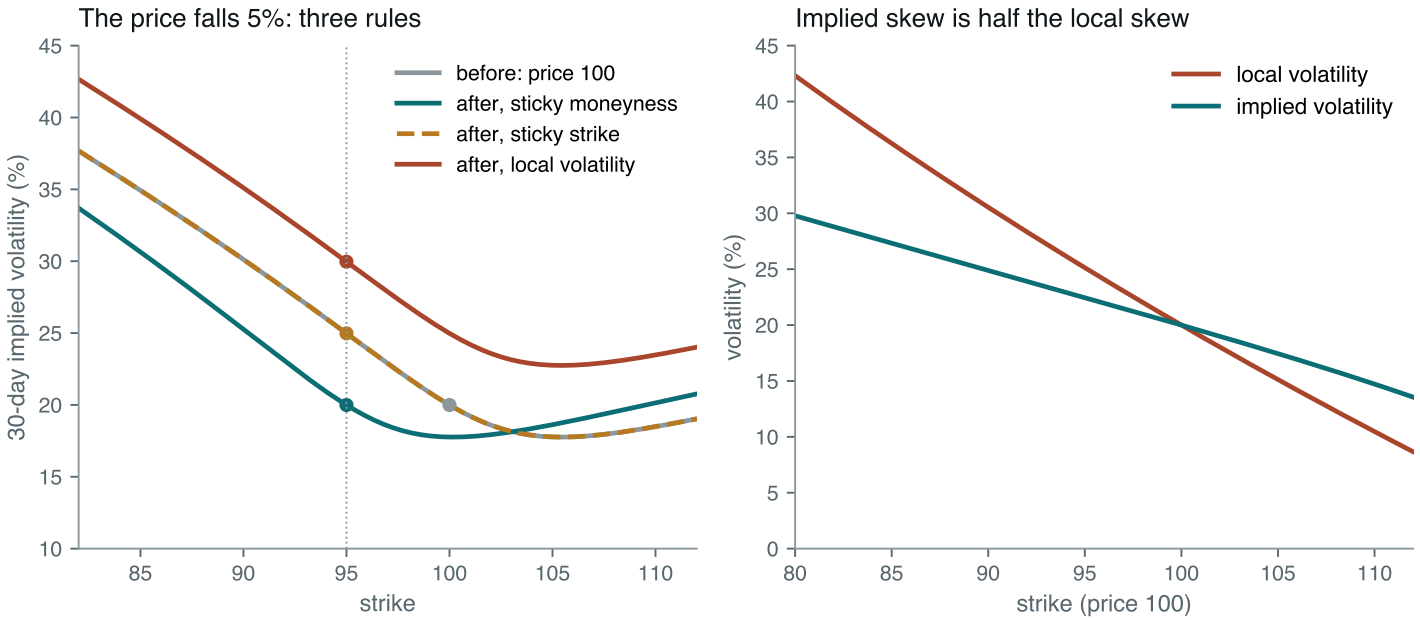
Local volatility is not a claim that volatility really is a function of the price. It is the unique diffusion consistent with today's prices, and it is genuinely useful for exactly that reason: it prices path-dependent payoffs consistently with the vanilla market, and it gives a well-defined answer to "what variance does the surface price at this price level and date".

8.2 Implied volatility averages local volatility

Combine Dupire with the bridge picture of §4.4 and a slogan falls out: an option's implied variance is roughly the average of local variance along the Brownian bridge from today's price to its strike (Gatheral, 2006). In the short-maturity limit this becomes exact and takes a memorable form (Berestycki, Busca and Florent, 2002): implied volatility at log-moneyness k is the **harmonic mean** of local volatility between the forward and the strike,

$$\frac{1}{\sigma_i(k)} \;=\; \frac{1}{k}\int_0^k \frac{dy}{\sigma_{\text{loc}}(y)}.$$

The immediate consequence is the practitioner's **rule of two**: near the money, the implied skew is about *half* the local skew, because the implied volatility at a strike averages local volatility over the whole interval from the forward to that strike, and the average of a linear function over an interval is its midpoint value.



The right panel makes the rule concrete. A local volatility that falls linearly from 42.3% at a strike of 80 to 10.5% at 110 generates implied volatilities of 29.8% and 14.7% at those strikes: the same 20% at the money, and roughly half the slope. [Simulated] Two practical implications follow. A local-volatility surface calibrated to a given implied skew is about twice as steep as that skew — which is why local volatility produces such violent behaviour in the wings. And an implied skew tells you about volatility *in the region between spot and strike*, not at the strike.

8.3 Dupire's formula joins the rate and the shape

Written in the surface's natural coordinates — total implied variance $w(k,T)$ against log-moneyness — Dupire's formula becomes the bridge promised in §1.4 (Gatheral, 2006):

$$\sigma^2_{\text{loc}}(k,T) \;=\; \frac{\partial w/\partial T}{g(k)}, \qquad g(k) = \left(1 - \frac{k\,\partial_k w}{2w}\right)^2 - \frac{(\partial_k w)^2}{4}\left(\frac{1}{w} + \frac{1}{4}\right) + \frac{\partial^2_k w}{2}.$$

Everything in this document is in that equation. The numerator is the **rate** direction: how fast total variance accumulates with expiry, which is forward variance at that moneyness (§5.4). The denominator is the **shape** direction: $g(k)$ is proportional to the risk-neutral density, via $f(k) = g(k)\,e^{-d_2^2/2}/\sqrt{2\pi w}$ (Gatheral and Jacquier, 2014).

That gives the cleanest possible statement of what a valid surface is. Local variance must be non-negative and finite, so:

Condition	Meaning	Violation is
$\partial w/\partial T \geq 0$ at fixed k	Total variance never decreases with expiry	Calendar-spread arbitrage
$g(k) > 0$ for all k	The implied density is positive	Butterfly arbitrage
$w(k)/\lvert k\rvert \leq 2$ asymptotically	Wings grow no faster than linearly	Lee-bound violation (§7.5)

An arbitrage-free surface is exactly one on which Dupire's formula returns a sensible local variance everywhere — and conversely, the fastest way to find the arbitrage in a fitted surface is to compute local variance and look for the places where it goes negative or explodes (§10.6).

8.4 What generates a smile: stochastic volatility and jumps

Local volatility fits any surface, which means it explains nothing about *why* the surface has the shape it does. Two mechanisms do.

Stochastic volatility. Let volatility itself be random (Hull and White, 1987; Heston, 1993). Uncertainty about the average variance produces a symmetric smile through the convexity argument of §4.3, with curvature growing in the volatility of

volatility. Correlation between the price and its volatility tilts that smile: negative correlation, the empirical norm for equities, means high-variance paths tend to be down paths, so the weights of low-strike options sit where variance is high, and their implied volatilities rise. Skew is therefore roughly proportional to (correlation × volatility of volatility).

Jumps. Let the price be able to move discontinuously (Merton, 1976; Bates, 2000). Jumps matter most at short horizons, because a diffusion has almost no chance of travelling far in a day while a jump does. That difference is sharp enough to be a test: Carr and Wu (2003) show that out-of-the-money option prices decay at different rates as expiry approaches depending on whether the process is continuous, discontinuous, or both, and find both components in S&P 500 options.

The empirical discriminator between model families is the **term structure of the at-the-money skew**. Pure stochastic-volatility diffusions generate short-dated smiles that are far too flat compared with what trades. [Fact] Jumps steepen the short end. The "rough volatility" literature reproduces the observed steepening with a fractional-noise volatility process, predicting a power-law skew $\propto T^{-\alpha}$ with α near 0.4 (Gatheral, Jaisson and Rosenbaum, 2018). [Contested]: Guyon and El Amrani (2022) find that a power law fits well from about a month out but fails at the shortest maturities, where the observed skew does not blow up, and that a capped or time-shifted power law fits the whole term structure better with one extra parameter. My read: the power law is a good description over the maturities most people trade, and the disagreement is about the few days at the very short end — which happen to be exactly where the fastest-growing part of the options market now lives.

8.5 When the price moves: sticky rules

Everything so far is about a surface at one instant. For anyone hedging, the harder question is what the surface does *next*, because an option's delta depends on it.

Write the smile as a function of strike, and ask what happens when the underlying falls 5%. Three conventional answers (Derman, 1999), all visible in the left panel of the figure above: [Simulated]

Rule	What stays fixed	At-the-money vol after a 5% fall	Skew-stickiness ratio
Sticky moneyness (sticky delta)	Volatility as a function of K/S	20.0% (unchanged)	0
Sticky strike	Volatility at each fixed strike	25.0% (+5.0 points)	1
Local volatility ("sticky implied tree")	The local volatility function	30.0% (+10.0 points)	2

The **skew-stickiness ratio** $\mathcal{R}$ (Bergomi, 2016) puts these on one scale: it is the move in at-the-money volatility per unit of log price move, divided by the at-the-money skew. Sticky moneyness is $\mathcal{R} = 0$, sticky strike is 1, local volatility is 2. Empirically, equity index surfaces sit between 1 and 2 at short maturities. [Practice] Derman (1999) observed that the *regime* changes: calm, trending markets behave more like sticky strike, frightened ones more like the local-volatility rule.

This is not a modelling nicety, because the rule determines the delta. Under a general rule, the fixed-strike implied volatility moves by $(\mathcal{R} - 1)$ times the skew for each unit of log price move, so the smile-adjusted delta is

$$\Delta = \Delta^{\mathrm{BS}} + \nu \, \frac{\partial \sigma_i}{\partial S}.$$

With a negative skew and $\mathcal{R} > 1$, fixed-strike volatility *falls* when the price rises, so the correct delta for a call is **below** the Black–Scholes delta — which is what Hull and White (2017) find empirically when they estimate the minimum-variance delta for equity index options. It also explains the practical complaint about local-volatility models made by Hagan, Kumar, Lesniewski and Woodward (2002) when they introduced SABR: a local-volatility model fitted to today's smile predicts that the smile moves in the *opposite* direction to the underlying, which is the reverse of what is observed, and it produces unstable hedges as a result.

8.6 How surfaces actually move

Four empirical regularities, useful both for hedging and for building features.

- **A few factors explain most of it.** Daily changes in the surface are dominated by a level factor, followed by a slope or term factor and a curvature factor (Cont and da Fonseca, 2002). Modelling the surface as level × term structure × one shape (as in §10.3's rescaling) is not a bad first approximation. [Fact]

- **Level moves are strongly negatively correlated with the underlying.** For equity indices this is the leverage effect seen in implied rather than realised volatility, and it is pronounced for both continuous and jump moves (Andersen, Bondarenko and Gonzalez-Perez, 2015). It is also asymmetric: volatility rises more on down moves than it falls on up moves of the same size.
- **Volatility mean-reverts, and the term structure encodes the speed.** Short-dated implied volatility is far more variable than long-dated; Stein (1989) argued that long-dated options nonetheless *over*react to short-dated shocks. [Contested], but the qualitative pattern — shocks decay along the term structure — is not in dispute.
- **Scheduled events reprice discontinuously.** The event lump of §5.5 enters the surface when the event enters the option's window and leaves the instant it is resolved.

§8 Key takeaways

1. Local volatility is the unique diffusion consistent with an entire surface: the value of one more instant of time at a strike, per unit of probability of being there.
2. Implied volatility is approximately an average of local volatility along the bridge from spot to strike, and exactly its harmonic mean in the short-maturity limit.
3. Hence the rule of two: local skew is about twice implied skew near the money.
4. Dupire's formula joins the two readings — forward variance over density — and the no-arbitrage conditions are exactly that both are well behaved.
5. Stochastic volatility generates curvature through uncertainty about variance and skew through spot–volatility correlation; jumps generate the steep short end.
6. The sticky rule you assume determines your delta. Empirical index surfaces sit between sticky strike and local volatility, so the right call delta is below the Black–Scholes one.
7. Surfaces move mostly in level, with slope and curvature second; level moves are strongly and asymmetrically negatively correlated with the underlying.

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9. Inverting one price

9.1 The problem, stated precisely

Given a market price V, a forward F, a discount factor D, a strike K and a time to expiry τ, find $\sigma > 0$ such that $\mathrm{BS}(F, K, \tau, D, \sigma) = V$.

Three preliminaries make the rest easy and are skipped surprisingly often.

- **Work with the out-of-the-money option.** Convert a put to a call (or back) with put–call parity, and invert whichever is out of the money. In-the-money options carry intrinsic value that swamps the time value, so the same absolute price error becomes a larger relative error in the quantity that actually depends on volatility.
- **Check the no-arbitrage bounds first.** If $V \le D(F-K)^+$ or $V \ge DF$, there is no implied volatility. Return a missing value; do not clamp to a boundary, because a clamped 500% or 0.1% will silently pollute every average downstream.
- **Reduce to two variables.** As §2.2 showed, the normalised price depends only on $k = \ln(K/F)$ and total volatility $s = \sigma\sqrt{\tau}$. Solve for s, then divide by $\sqrt{\tau}$. This keeps the root-finder's scale independent of maturity.

9.2 Algorithms

Bisection always works: price is monotone in volatility, so a bracket of, say, $[10^{-6}, 10]$ can be halved about fifty-five times to reach machine precision. It is slow but unconditionally safe, and it is the right fallback inside any production routine.

Newton–Raphson uses vega, $\sigma_{n+1} = \sigma_n - (\mathrm{BS}(\sigma_n) - V)/\nu(\sigma_n)$, and near the money it is extraordinarily fast. Start from the at-the-money shortcut of §2.4 — $\sigma_0 = V/(0.4\,DF\sqrt{\tau})$ — and the running example converges in **one step**: 19.944% goes to 20.000000%. [Simulated] Away from the money, closed-form approximations that extend the shortcut, such as Corrado and Miller (1996), make better starting values, though their errors grow with distance from the forward.

Away from the money it can fail badly, and the failure is instructive. Take the 110 call with 30 days left, priced at 25% volatility (0.321):

Starting guess	Newton iterates
20%	26.66% → 25.08% → 25.0002% → 25.000000%
Manaster–Koehler 152.3%	36.47% → 26.90% → 25.11% → 25.0004% → 25.000000%
10%	681% → −196% → diverged
5%	vega is 3×10^{-9}; the step is numerically undefined

The cure is a starting value that cannot overshoot. Manaster and Koehler (1982) showed that starting at the volatility where the price is an inflection point in σ,

$$\sigma_0 = \sqrt{\frac{2\,|\ln(F/K)|}{\tau}},$$

makes Newton converge monotonically from above for any valid price — 152.3% for this option, then down to the answer without oscillation. Below that point the price is convex in volatility and above it concave, which is exactly the condition under which Newton behaves.

Production libraries do something better. Jäckel (2015) constructs rational-function initial guesses tuned to the log-moneyness region, then applies two Householder iterations of order four, reaching full double precision for every input in at most two steps. It is the de facto standard, and open-source implementations exist. [Practice] Unless you have a reason not to, use one of them; if you write your own, use Newton from the Manaster–Koehler start with a bisection safeguard, and test it on deep wings and on options with hours to expiry.

For **American** options there is no closed form to invert. Either price with a binomial or trinomial tree inside the root-finder, or "de-Americanise" first: compute the early exercise premium on a tree at a trial volatility, subtract it to get a European-equivalent price, invert Black–Scholes, and iterate once or twice. [Practice]

9.3 Conditioning: when a cent is a volatility point

Inversion error is price error divided by vega:

$$\Delta\sigma_i \approx \frac{\Delta V}{\nu}.$$

Since vega collapses in the wings and at short maturities (§2.3), so does the information content of a quoted price. The right panel of the §2.3 figure plots this; here it is as numbers, for a five-cent price error at a flat 20% volatility: [Simulated]

Strike	7 days	30 days	180 days
90	1,324 vol pts	2.49 vol pts	0.25 vol pts
95	5.16	0.67	0.20
100	0.91	0.44	0.18
105	4.17	0.61	0.19
110	322	1.66	0.21
120	—	62.6	0.38

At 30 days the at-the-money implied volatility is accurate to half a point on a five-cent error, while the 120-strike number is noise: that option is worth about a tenth of a cent at this volatility, so the "error" exceeds the price. At seven days the usable range has narrowed to roughly 95–105. Hentschel (2003) makes this point formally and proposes weighted estimators; the practical versions are:

- **Weight by vega** (or equivalently by the inverse of the implied-volatility bid–ask spread) when fitting a smile, so the wings cannot drag the fit.
- **Quote the uncertainty.** Convert the bid–ask spread into volatility: $\sigma_i^{\text{ask}} - \sigma_i^{\text{bid}} \approx (\text{ask} - \text{bid})/\nu$. An option whose volatility bid–ask is eight points wide should not be a data point.
- **Drop options priced within a tick or two of zero.** Their implied volatilities are determined by tick size, not by the market.

9.4 The inputs matter more than the algorithm

Having spent a section on root-finders, here is the order of importance in practice, which is nearly the reverse of the order in which people usually worry about it.

1. The forward (dividends, borrow, rates). This dominates everything. Put–call parity gives it for free: $C - P = D(F - K)$, so regressing the call-minus-put price on the strike across near-the-money pairs yields $-D$ as the slope and DF as the intercept. Use the options' own implied forward rather than a spot price plus a dividend assumption.

How badly does it matter? Take the running example and misestimate the forward by half a point — 99.5 instead of 100, a 0.5% error, entirely plausible from a wrong dividend or a hard-to-borrow stock. Inverting the *correct* market prices with the *wrong* forward gives a call implied volatility of 22.17% and a put implied volatility of 17.77%. [Simulated] **A 0.5% forward error manufactures a 4.4-point call–put volatility gap** — a completely fictitious skew, and a large one. This is also the best diagnostic available: if your call and put implied volatilities disagree at the same strike, the forward is wrong before anything else is.

2. Synchronisation. The option quote and the underlying price must be from the same instant. Historical datasets where options close at 4:15pm and the underlying at 4:00pm build a systematic error into every number. Using the implied forward sidesteps this almost entirely, which is another reason to prefer it.

3. The clock. Time to expiry is measured in years, but which year? Calendar days over 365 spread variance evenly over days the market is shut; trading days over 252 assume weekends carry none. For a 30-day option the difference is under a percent. For an option with six hours to live it is enormous: six hours is 0.000685 calendar-years but 0.003663 trading-years, so **the same price implies a volatility 2.3 times higher on the calendar clock than on the trading clock**. Any comparison of intraday or zero-days-to-expiry implied volatilities has to state its clock. Desks handle the general case with a variance clock that weights each hour by its expected variance share. [Practice]

4. Exercise style. Listed single-stock options in the US are American. The early exercise premium is largest for in-the-money puts and for calls facing a dividend, so inverting them with a European formula overstates volatility. Using out-of-the-money options removes most of the problem at no cost.

5. Which price. Mid-quote is the default; last trade is stale and sometimes hours old. Wide markets should be down-weighted, not averaged into the fit as if they were equally informative.

6. Discounting. The rate matters only through D and the forward, and for short-dated options it is a rounding error — which is why it is safe to leave it last.

§9 Key takeaways

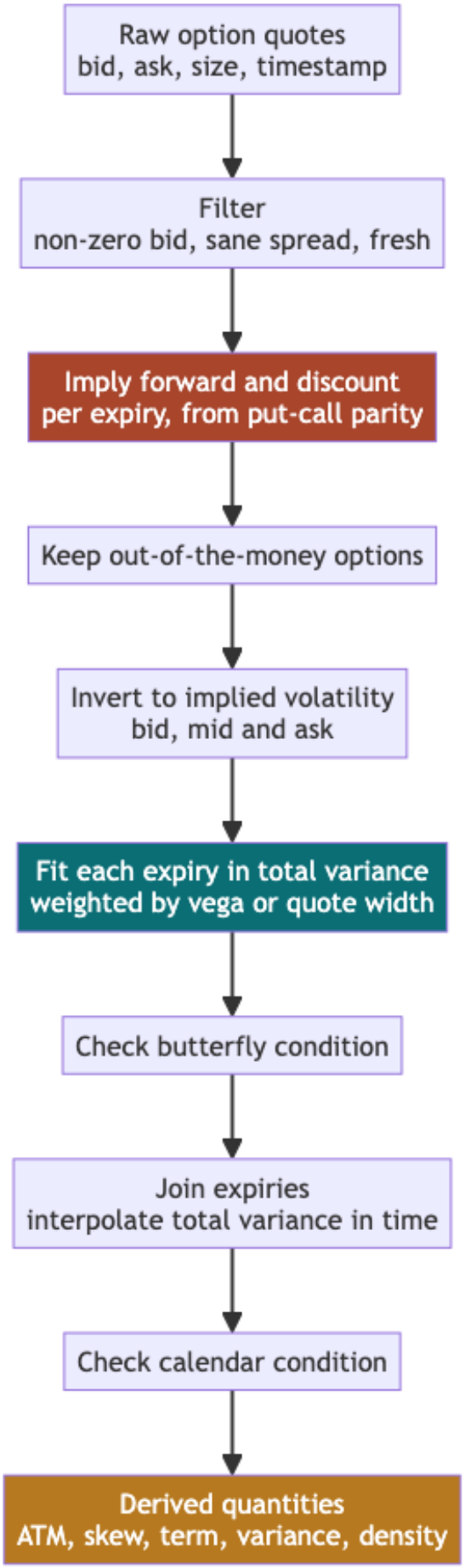
1. Invert the out-of-the-money option, check the no-arbitrage bounds, and solve in total volatility $\sigma\sqrt{\tau}$.
2. Newton from the at-the-money shortcut converges in one step near the money and diverges in the wings; the Manaster–Koehler inflection start fixes that.
3. Use a proven implementation (Jäckel's is the standard) rather than writing your own unless you have a reason.
4. Implied-volatility error is price error over vega, so wing and short-dated numbers are noise: a five-cent error is half a point at the money and hundreds of points in a seven-day wing.
5. Weight fits by vega, convert bid–ask spreads into volatility, and drop options priced at a tick.
6. The forward matters more than everything else combined: a 0.5% error manufactures a 4.4-point fake skew. Imply it from put–call parity.
7. Disagreement between call and put implied volatility at the same strike is a diagnostic, not a trading signal.

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10. From quotes to a surface

10.1 The pipeline

Building a surface is a data-engineering problem with a small amount of mathematics at the end. The order of operations matters, because errors early on are impossible to detect later.


Raw option quotes
bid, ask, size, timestamp
Filter
non-zero bid, sane spread, fresh
Imply forward and discount
per expiry, from put-call parity
Keep out-of-the-money options
Invert to implied volatility
bid, mid and ask
Fit each expiry in total variance
weighted by vega or quote width
Check butterfly condition
Join expiries
interpolate total variance in time
Check calendar condition
Derived quantities
ATM, skew, term, variance, density

10.2 Cleaning quotes

The filters below are conventional, and each one exists because of a specific way the data lies. [Practice]

Filter	Why
Drop zero or missing bids	The implied volatility is undefined or tick-determined (§9.3)
Drop crossed or locked markets	Stale or erroneous feed data
Drop quotes wider than a threshold in volatility terms	The midpoint carries almost no information
Keep out-of-the-money options only	Avoids American exercise premium and intrinsic-value dominance
Require a recent timestamp	An option that has not been quoted for an hour is not a price
Drop options priced within a tick or two of zero	Tick size, not volatility
Flag outliers against a first-pass fit rather than deleting them blind	Real dislocations exist and are sometimes the point

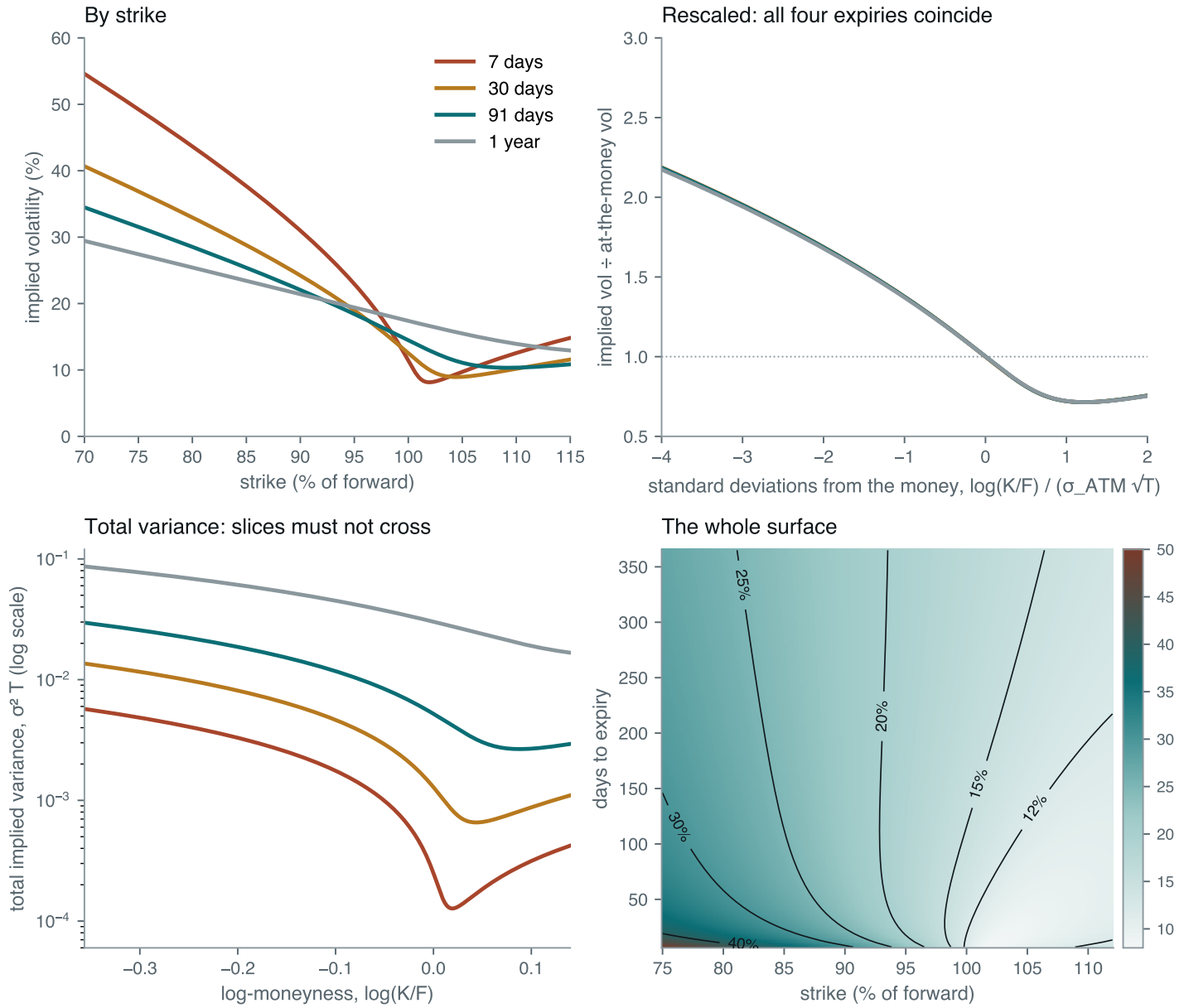
Keep the bid and ask implied volatilities alongside the mid. They are the natural weights for fitting and the natural error bars for any picture (§12.1).

10.3 Coordinates

The choice of horizontal axis changes what a surface looks like and what can be compared.

Coordinate	Definition	Good for	Watch out
Strike	K	Trading a specific contract	Nothing is comparable across expiries or dates
Moneyness	K/F	Simple comparisons	Ignores that a 10% move means more at a year than a week
Log-moneyness	$k = \ln(K/F)$	The natural coordinate for the maths	Still maturity-dependent
Standardised moneyness	$k/(\sigma_{\text{ATM}}\sqrt{T})$	Comparing shapes across expiries and regimes	Depends on the volatility level you divide by
Delta	$N(d_1)$	FX convention, vendor surfaces	The strike behind a 25-delta point moves with volatility

The payoff from the right coordinate is visible in the figure below. The top-left panel is the same SSVI surface plotted against strike: the seven-day slice looks violently steep and the one-year slice almost flat. The top-right panel divides each slice by its own at-the-money volatility and plots it against standard deviations from the money — and all four expiries collapse onto a single curve. [Simulated] A surface really is, to a good approximation, **a level, a term structure, and one shape**.



The delta axis deserves its own warning. A 25-delta put is a fixed *probability-like* distance from the money, so the strike it refers to moves further away as volatility rises. Comparing "25-delta skew" across a calm period and a crisis therefore mixes a change in shape with a change in which strikes are being compared. [Practice]

10.4 Fitting one expiry

Fit in total variance w against log-moneyness k, weighted by vega or by the inverse of the quote width in volatility terms.

- **Polynomials in moneyness.** The "practitioner Black–Scholes" approach: fit a quadratic to implied volatilities and use it as an interpolator. Dumas, Fleming and Whaley (1998) found that this ad hoc smoothing predicted and hedged at least as well as the deterministic local-volatility models of its day, which was an uncomfortable result at the time and remains a useful benchmark. Its fatal flaw is extrapolation: a quadratic in k violates Lee's linear bound (§7.5), so densities computed from it go negative in the wings.
- **SVI.** The standard parametric smile, introduced by Gatheral at a 2004 conference and given explicit no-arbitrage conditions by Gatheral and Jacquier (2014):

$$w(k) = a + b\left[\rho\,(k-m) + \sqrt{(k-m)^2 + s^2}\right].$$

Five parameters with clean readings: a the overall level, b the angle between the wings, $\rho \in (-1, 1)$ the rotation that produces skew, m the horizontal translation, s the smoothness at the money. The wings are linear by construction, which is what makes it Lee-consistent: the resulting wing-slope bound is exactly $b(1 + |\rho|) \le 2$ (§7.5). That bound

is necessary for an arbitrage-free fit but not sufficient by itself — it constrains the wings alone, and a calibration can satisfy it while still producing butterfly arbitrage nearer the money unless s is also large enough relative to b and ρ.

- **SSVI.** A surface-level version in which every slice is determined by the at-the-money total variance $\theta(T)$ and a single shape function $\phi(\theta)$ — typically a power law. It is arbitrage-free by construction under simple parameter conditions, and it is what generated the figure above. Its assumption (one shape, rescaled) is exactly the empirical regularity of §10.3.
- **SABR.** The rates and FX standard (Hagan, Kumar, Lesniewski and Woodward, 2002): a stochastic-volatility model with four parameters — level α, backbone β (0 for normal, 1 for lognormal), correlation ρ for skew, volatility-of-volatility ξ for curvature (written ν in the original paper, which would collide with vega here) — and a closed-form implied-volatility approximation. (Two of SABR's own letters are reused elsewhere in this document for unrelated quantities: α is also the rough-volatility exponent of §8.4, and β is also Lee's wing exponent $\beta_{R,L}$ of §7.5.) The asymptotic formula can imply negative densities at low strikes and long maturities, which later arbitrage-free versions address. [Practice]
- **The FX three-point convention.** Foreign-exchange desks quote a smile as three numbers per expiry: at-the-money (usually the delta-neutral straddle), the 25-delta **risk reversal** $\sigma(25\Delta\mathrm{c}) - \sigma(25\Delta\mathrm{p})$, and the 25-delta **butterfly** $\frac{1}{2}[\sigma(25\Delta\mathrm{c}) + \sigma(25\Delta\mathrm{p})] - \sigma_{\mathrm{ATM}}$, often with 10-delta points as well (Reiswich and Wystup, 2010). For the stylised surface above, the 30-day risk reversal is -6.6 volatility points and the butterfly $+0.8$. [Simulated]

What actually matters in a production fit is less the family than four properties: the fit lies inside the bid–ask across the liquid strikes; it is arbitrage-free; its parameters are stable from day to day (unstable parameters mean unstable hedges); and it extrapolates sensibly beyond the quoted strikes.

10.5 Joining expiries

Interpolate **linearly in total variance at fixed moneyness**. This is what the VIX does between its two bracketing expiries (§5.2), and it is the only interpolation that keeps forward variance constant between quoted dates rather than manufacturing humps. Interpolating implied volatility directly against time, by contrast, implies wobbling forward variances and can even produce negative ones.

Two details. Interpolate at fixed *forward* moneyness, not fixed strike, so that the forward's drift between expiries does not leak into the shape. And beyond the last listed expiry, extrapolate with flat forward variance rather than flat volatility. [Practice]

10.6 No-arbitrage checks

Run these on every surface, every day. They are cheap, and each one catches a class of error that silently corrupts everything downstream.

Check	Condition	Catches
Calendar	$w(k,T)$ non-decreasing in T at fixed k	Bad term interpolation, stale expiries, wrong forwards
Butterfly	$g(k) > 0$ for all k (§8.3); equivalently call prices convex in K	Overfitted smiles, negative densities
Wings	$w(k)/\lvert k\rvert \le 2$ asymptotically	Polynomial fits, reckless extrapolation
Digitals	$-\partial C/\partial K \in [0, D]$	Non-monotone price curves
Parity	Call and put implied volatilities agree at each strike	Wrong forward (§9.4)

The stylised SSVI surface above passes all of them: its minimum butterfly value is $g = 0.25$ and total variance is strictly increasing in expiry everywhere on the grid. [Simulated] The reason to care is that every derived quantity in §11 involves differentiating or integrating the surface, and differentiation amplifies exactly the small violations that eyeballing a plot will never reveal.

§10 Key takeaways

1. Building a surface is mostly data work: filter, imply the forward, invert, fit, check, join, check again.
2. Keep bid and ask implied volatilities; they are the fit weights and the error bars.
3. Plotted against strike, short expiries look steep for the wrong reason; rescaled by at-the-money volatility and plotted in standard deviations, a whole surface is nearly one shape.

4. Fit in total variance against log-moneyness. SVI and SSVI are the modern defaults; polynomial fits violate the wing bound and produce negative densities.
5. A 25-delta point moves in strike terms as volatility changes, so delta-based skew measures mix level and shape.
6. Interpolate between expiries linearly in total variance; anything else invents forward-variance structure.
7. Check calendar, butterfly, wing, digital and parity conditions daily — derivatives of the surface amplify violations you cannot see by eye.

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11. Derived quantities

A surface is rarely the end product. What gets stored, plotted, traded and fed into models is a handful of numbers extracted from it. Each has conventions that differ between desks, and mixing conventions is one of the most common sources of nonsense.

11.1 Which at-the-money?

"At-the-money implied volatility" names at least five different numbers.

Definition	Strike	Notes
At-the-money spot	$K = S$	Common in equity data; drifts with rates and dividends
At-the-money forward	$K = F$	The natural choice; $k = 0$
Delta-neutral straddle	$d_1 = 0$, i.e. $k = \frac{1}{2}\sigma^2\tau$	The FX convention
50-delta call	$N(d_1) = 0.5$	Same as the delta-neutral straddle for unadjusted deltas
Nearest listed strike	whatever trades	Introduces a sawtooth as the price moves between strikes

The gaps are small for short-dated, low-volatility options and material otherwise. The delta-neutral straddle sits at $k = \frac{1}{2}\sigma^2\tau$ above the forward: 0.16% for a 30-day option at 20% volatility, but 2% for a one-year option. With a one-year skew of 0.4 volatility points per 1% of strike (§11.3), that is a 0.8-point difference between two things both called "the at-the-money volatility". Always record which one you mean.

11.2 Constant-maturity series

To compare today's 30-day volatility with last year's, you need a constant-maturity series, which means interpolating between listed expiries — again linearly in total variance (§10.5), at fixed moneyness, as the VIX does. Two failure modes to avoid: rolling from one listed expiry to the next produces a sawtooth that looks like signal and is not; and letting the front expiry get within a few days of expiration imports the noise of §9.3 into the series.

11.3 Skew, curvature and term-structure measures

Measure	Definition	Captures	Pitfall
At-the-money slope	$\partial\sigma_i/\partial k$ at $k=0$	Skew, locally	Sensitive to the fit near the money
Fixed-moneyness spread	$\sigma_i(90\%)-\sigma_i(100\%)$	Skew over a fixed range	Means different things at different maturities
25-delta risk reversal	$\sigma_i(25\Delta\text{c})-\sigma_i(25\Delta\text{p})$	Skew, FX convention	Strikes move with the volatility level
25-delta butterfly	$\frac{1}{2}[\sigma_i(25\Delta\text{c})+\sigma_i(25\Delta\text{p})]-\sigma_{\text{ATM}}$	Curvature, fat tails	Same
Normalised skew	Slope $\times\sqrt{T}$, or risk reversal over at-the-money vol	Shape free of level and maturity	The right one for time series and cross-sections
Risk-neutral skewness	From the moment strips of §7.3	Whole-distribution asymmetry	Needs the wings, hence extrapolation
Term slope	$\sigma_i(3\text{M})-\sigma_i(1\text{M})$, or a ratio such as VIX to VIX3M, Cboe's three-month counterpart	Where the market thinks volatility is heading, plus term premia	Inverts in stress, which is the signal

Two of these are worth dwelling on. **Normalisation matters**: raw skew measures rise mechanically when the volatility level rises and shrink with maturity roughly as $T^{-0.4}$ to $T^{-0.5}$ (§8.4), so an unnormalised skew time series is mostly a disguised volatility time series. On the stylised surface, the at-the-money skew falls from 2.85 volatility points per 1% of strike at a week to 0.40 at a year, with a fitted exponent of -0.50. [Simulated] And the **term slope** is the single most useful summary of the curve: its inversion is a reliable marker of stress, and it carries information about the variance risk premium (Johnson, 2017).

11.4 Model-free implied variance in practice

To compute a VIX-style number on an arbitrary underlying, follow §5.2 but make three choices deliberately.

- **Truncation versus extrapolation.** Cboe truncates after two consecutive zero bids, which is reproducible but makes the index's strike coverage vary with market conditions (Andersen, Bondarenko and Gonzalez-Perez, 2015). Integrating a fitted, Lee-consistent smile out to a fixed number of standard deviations instead gives a smoother series at the cost of a model in the tails. For research I prefer the fitted version and would report both.
- **Interpolation grid.** Sum over a fine strike grid from the fitted smile rather than over listed strikes; single-stock strike spacing is coarse enough to bias the sum.
- **Validation.** Compute the VIX itself with Cboe's exact rules and check you reproduce the published value to a few hundredths. Nothing else tests an implied-volatility pipeline as thoroughly, because the calculation touches the forward, the filters, the clock and the interpolation all at once. [Practice]

A related quantity worth knowing: weighting the same option strip by $1/F^2$ instead of $1/K^2$ gives the risk-neutral variance of the *simple* return, which Martin (2017) calls SVIX and shows is a lower bound on the equity risk premium under a weak condition. It is the most direct route from option prices to an expected return.

11.5 Densities and moments

Recover the density from the **fitted** smile, never by differencing raw prices twice — market noise differentiates into garbage. Convenient forms:

$$f(k)=\frac{g(k)}{\sqrt{2\pi w(k)}}\,e^{-d_2^2/2},$$

with $g(k)$ the butterfly function of §8.3 (Gatheral and Jacquier, 2014). Malz (2014) gives a robust recipe that interpolates implied volatility in delta space with a clamped spline before differentiating; Jackwerth and Rubinstein (1996) give the smoothness-maximising alternative. The tails beyond the quoted strikes are extrapolation, and every published density is partly an assumption — a point central banks publishing these series have always been careful to make (Bank of England published such densities weekly until 2021; the Minneapolis Fed publishes market-based probabilities today).

11.6 Forward volatility and event moves

Forward variance between two expiries is the difference of total variances (§5.4), and the implied event move comes from the same arithmetic with a dated lump (§5.5). One warning that catches people: **forward implied volatility is not the implied volatility of a forward-starting option.** The former is a statement about the average variance between two dates; the latter also depends on the forward smile and the volatility of volatility, and trades at a different number.

11.7 Implied correlation

Index variance and constituent variances are linked by an accounting identity, so the correlation the market prices can be backed out. Give stock j an index weight ω_j and an implied volatility σ_j, write σ_{idx} for the index's implied volatility, and assume a single correlation between every pair of stocks:

$$\rho_{\text{imp}} = \frac{\sigma_{\text{idx}}^2 - \sum_j \omega_j^2 \sigma_j^2}{\sum_{j \neq l} \omega_j \omega_l \sigma_j \sigma_l}.$$

An index at 18% built from 50 equally weighted stocks each at 30% implies a correlation of **0.35**. Use variance-swap volatilities rather than at-the-money ones on both sides if you can, since the identity is about variance. Implied correlation is the quantity a dispersion trade is long or short (index volatility against constituent volatility), and it carries its own risk premium: index options are expensive relative to single-stock options precisely because correlation risk is priced (Driessen, Maenhout and Vilkov, 2009).

11.8 Rank, percentile, and the implied–realised spread

Implied-volatility rank is $(\sigma_i - \min)/(\max - \min)$ over a lookback window; **implied-volatility percentile** is the fraction of days in the window below today's level. They are routinely treated as synonyms and routinely disagree. In the simulated year in the figure of §12.6, implied volatility is 18.8% today, the year's low is 14.5% and its high 54.7%: rank says **11%** and percentile says **56%**. [Simulated] One three-week spike set the maximum and now dominates the rank; the percentile is unmoved by it. Percentile is the more robust statistic, and neither one knows anything about the current regime or about the premium.

For the **implied–realised spread**, decide what you are measuring before you compute it:

- *Ex post*, for measuring the premium: today's 30-day implied volatility against the realised volatility of the following 30 calendar days. This requires waiting, and it cannot be used as a signal without look-ahead.
- *Ex ante*, as a signal: today's implied volatility against **trailing** realised volatility, which is what Goyal and Saretto (2009) use to predict delta-hedged option returns in the cross-section of stocks.

In both cases, match the horizon (30 calendar days is about 21 trading days), work in variance and convert at the end, and remember §4.5: for a single option the theoretically right comparison is dollar-gamma-weighted realised variance, not a plain estimator.

§11 Key takeaways

1. "At-the-money" names at least five different strikes; the delta-neutral straddle sits $\frac{1}{2}\sigma^2\tau$ above the forward, worth 0.8 volatility points at a year.
2. Constant-maturity series require interpolation in total variance; rolling between listed expiries manufactures a sawtooth.
3. Normalise skew measures by level and maturity, or the series is a disguised volatility series.
4. The term-structure slope is the most informative single summary of the curve.
5. Validate a whole pipeline by reproducing the published VIX to a few hundredths.
6. Densities come from fitted smiles, and their tails are always partly assumption.
7. Forward implied volatility is not the volatility of a forward-starting option.
8. Implied-volatility rank and percentile can disagree by 45 points; prefer percentile, and match horizons whenever comparing implied with realised.

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12. Pictures of implied volatility

Implied volatility is one of the few quantities in finance that practitioners genuinely look at rather than only compute. This section takes the standard pictures in turn and says what each one answers and how each one misleads. The figures themselves have already appeared where their concepts were introduced; the point here is the reading.

12.1 The smile, and the choice of axis

What it shows. One expiry: implied volatility against strike, log-moneyness, delta or standardised moneyness (§10.3).

How to read it. Level is the market's price for variance near the money; the tilt is the price of asymmetry; the curl is the price of tails. Overlay the bid and ask implied volatilities as a band: it shows instantly which part of the smile is information and which is an artefact of the quote width (§9.3).

How it misleads. A strike axis makes short-dated smiles look dramatic and long-dated smiles look flat, most of which is the $\sqrt{T}$ scaling rather than a change of shape (§10.3). A delta axis quietly changes which strikes are being compared as the level moves.

12.2 The term structure

What it shows. At-the-money implied volatility against expiry (§5.4), ideally with the total-variance version beside it.

How to read it. Slope in total variance is forward variance. Contango means the market prices mean reversion upward; backwardation means it expects today's turbulence to fade; a hump means a dated event.

How it misleads. Reading levels rather than slopes hides the forward variance, which is where the information is. Comparing curves across dates without constant-maturity interpolation compares different maturities. And weekly expiries produce a weekend-shaped sawtooth in any curve plotted on a calendar clock (§9.4).

12.3 The whole surface

What it shows. Both dimensions at once, as a heat map with contours (the fourth panel of the §10.3 figure), as small multiples of slices, or as a 3D rendering.

How to read it. The heat map is the only one of the three in which a reader can actually retrieve a number; contours make the skew and term structure visible simultaneously. Small multiples are better for comparing shapes. 3D renderings look impressive and are nearly useless for reading values — they hide exactly the short-maturity wing region where the interesting things happen.

How it misleads. Any surface plot flatters the interpolation: the plotted grid is mostly points where no option trades.

A particularly useful variant is the **change** plot: today's surface minus yesterday's, which shows immediately whether the day's move was level, skew or term (§8.6).

12.4 The distribution behind the smile

What it shows. The risk-neutral density implied by one expiry's smile (§7.2), optionally against a lognormal fitted at the money.

How to read it. This is the picture that makes skew and kurtosis concrete, and the one to put in front of a non-specialist. Plot the tails on a log scale if the tails are the point.

How it misleads. Two ways, both serious. The tails beyond the quoted strikes are extrapolation (§11.5). And the density is risk-neutral: it overstates the real-world probability of the states people pay to hedge (§6.1). A density plot labelled "the market's view of where the index will be" is wrong twice over.

12.5 Time series

What it shows. A single number through time: a VIX-style index, an at-the-money series, a normalised skew, the term-structure ratio, or implied minus realised.

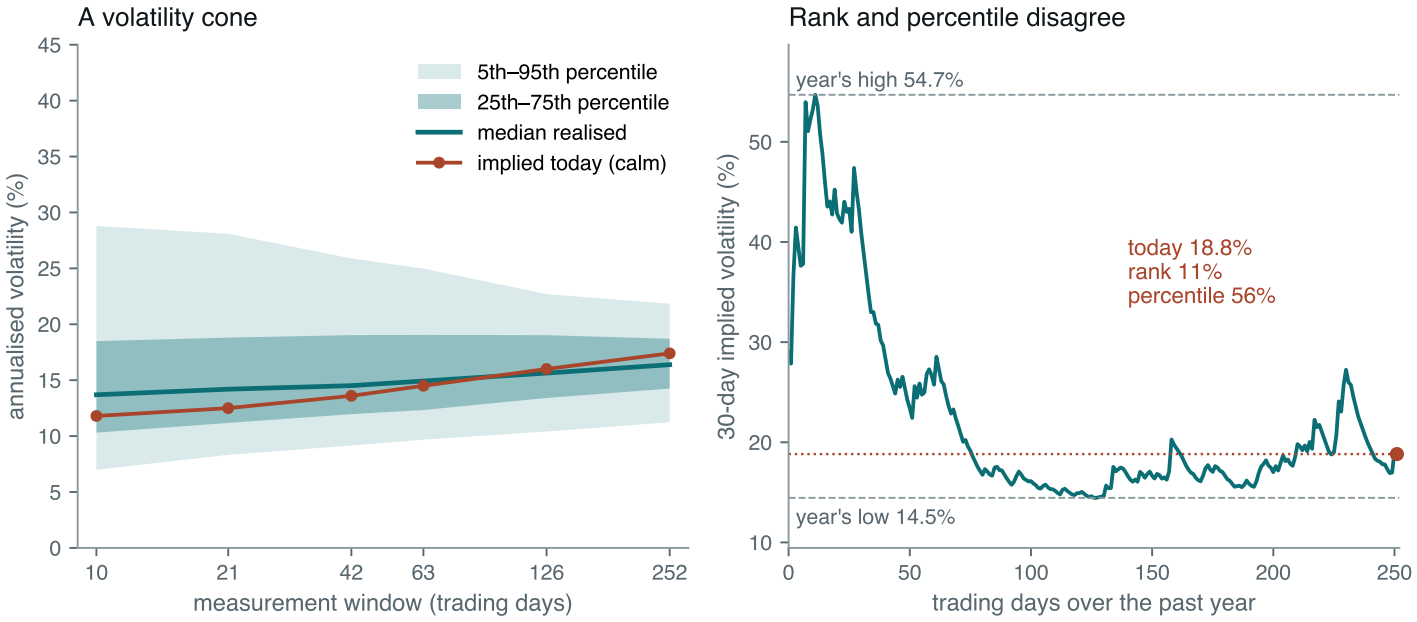
How to read it. Volatility series mean-revert, spike upward, and have strongly right-skewed distributions; a log scale is usually the honest choice. For the implied–realised comparison, plot realised volatility **forward-aligned** — the realised volatility of

the 30 days *after* each date, plotted at that date — so the picture answers “was implied volatility too high?” rather than “did volatility just happen?”.

How it misleads. The forward-alignment error above is the most common chart mistake in this field: plotting trailing realised volatility against implied volatility makes implied look like a lagging indicator of something it is actually forecasting. Overlapping windows also make both series far smoother than the underlying information warrants.

12.6 Cones, ranks and percentiles

What it shows. A **volatility cone** (Burghardt and Lane, 1990) plots percentiles of realised volatility by measurement horizon, with the current implied term structure laid over it.



How to read it. The cone’s narrowing is the square-root law made visible: short windows of realised volatility scatter widely, long ones average out. In the simulated example, the 5th-to-95th percentile range is 7.0–28.8% at ten days and 11.2–21.8% at a year. [Simulated] Laying the implied term structure over it shows where the market is pricing volatility relative to how that underlying has actually behaved.

How it misleads. Three ways. The cone is **unconditional**: it mixes calm and crisis history, so “below the median” does not mean cheap when today is calm. It compares implied with realised, which differ by the risk premium, so implied volatility sitting a few points above the median realised is the normal state of the world, not a sell signal (§6.3). And rank versus percentile can disagree violently, as the right panel shows.

12.7 The spot–volatility scatter

What it shows. Daily log returns of the underlying against daily changes in at-the-money implied volatility (or in a VIX-style index).

How to read it. The slope is the empirical spot–volatility beta, and dividing it by the at-the-money skew gives the skew-stickiness ratio of §8.5 — the number that decides whether your delta should be above or below the Black–Scholes one. Fit the two halves separately: volatility rises more on down days than it falls on up days.

How it misleads. Fixed-strike and fixed-moneyness volatility changes are different series, differing by the skew times the move; plotting one and interpreting it as the other is an easy way to mis-measure the ratio.

12.8 Which picture answers which question

Question	Picture	Look for
Is this option rich or cheap relative to its neighbours?	Smile with bid–ask band and fitted curve	Residual outside the band
What does the market price for variance to a horizon?	Term structure in total variance	Slope between the relevant dates
Is there an event, and how big?	Term structure, total-variance panel	A vertical step
What outcomes does the market price?	Implied density	Left-tail mass, mode location
Is volatility high for this asset?	Cone plus percentile of own history	Position versus conditional forecast, not versus median
Has the market's risk appetite changed today?	Surface-change plot; term-structure ratio	Level versus skew versus term; inversion
Should my delta be above or below Black–Scholes?	Spot–volatility scatter	Slope, and its asymmetry
Is my surface arbitrage-free?	Total variance against log-moneyness, all expiries	Crossing slices; non-convexity

§12 Key takeaways

1. The axis you plot a smile against determines what looks steep; rescaled coordinates separate shape from level and maturity.
2. Plot bid and ask implied volatilities as a band — most wing "structure" is quote width.
3. Read term structures in total variance, where the slope is forward variance and an event is a step.
4. Heat maps with contours beat 3D surface renderings for every practical purpose.
5. Implied densities are extrapolated in the tails and risk-neutral everywhere; label them as prices, not beliefs.
6. Align realised volatility *forward* when comparing it with implied, or you will conclude that implied volatility lags what it is forecasting.
7. Volatility cones are unconditional and ignore the premium; rank and percentile can disagree by 45 points on the same series.

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13. How practitioners use implied volatility

The same number serves very different purposes depending on who is holding it. This section goes user by user. Each use follows the same template: the question being asked, the implied-volatility quantity that answers it, the logic, and the way it fails.

13.1 Market makers

Quoting. A market maker does not price options one at a time; it runs a fitted surface and generates quotes from it, shading for inventory. The surface is the pricing engine, the quotes are its output, and a trade that arrives at a strike where the desk is already long volatility gets a worse bid (Dealer Hedging, section 9.2). *Fails when* the surface is stale in a fast market — the worst quotes a desk shows are the ones generated from yesterday's skew.

Marking and P&L attribution. Positions are marked to the surface and the day's P&L is split into delta, gamma, theta and vega (§3.2). *Fails when* the surface used for marking is the desk's own rather than a consensus one: a book can be marked into profit by moving the skew, which is why independent price verification exists. [Practice]

Choosing the hedge ratio. Delta depends on what you assume the surface does when the price moves (§8.5). Desks either assume a sticky rule or estimate the minimum-variance delta empirically (Hull and White, 2017). *Fails when* the assumed rule and the market's actual regime diverge — which Derman (1999) showed happens as markets move between calm and frightened states.

Which volatility to hedge at. Hedging with deltas from the implied surface makes daily P&L match the identity of §4.1 and keeps the book consistent with its marks, at the cost of path-dependent total P&L (Dealer Hedging, section 4.6).

13.2 Volatility traders

The core trade. Buy an option and hedge it if you think realised volatility will beat implied; sell and hedge if you think the reverse. Expected P&L is $\nu \times (\sigma_i - \sigma_r)$ — about $\$1.14$ per option per 10 volatility points in the running example — and the realised outcome scatters around it by the hedging error of §4.2. *Fails when* the forecast edge is smaller than the premium (§6.3), the hedging error, and costs. My rule of thumb: if the gap between implied volatility and a defensible forecast is not several volatility points, there is no trade.

Harvesting the variance risk premium. Short straddles, strangles, put-writing and short variance swaps all monetise the wedge of §6.3. The economics are sound and the distribution is brutal: many small gains, rare enormous losses, with the losses arriving exactly when everything else in the portfolio is also losing. Size it by stress loss, not by volatility of returns. *Fails when* leverage is set from realised Sharpe ratios computed in the calm period — the failure mode of February 2018.

Term-structure trades. Calendars express a view on forward variance rather than on volatility. The curve's slope predicts the returns to holding variance exposure (Johnson, 2017), and the VIX futures premium behaves in ways that are informative about risk pricing (Cheng, 2019). *Fails when* the trade is really a short-volatility carry position in disguise, which most steep-contango calendar trades are.

Skew trades. Risk reversals and put spreads take a view on the shape rather than the level. *Fails when* the position is unintentionally long or short the level too: a risk reversal has vega if the smile shifts, and a "pure skew" trade needs the level hedged deliberately.

Dispersion. Selling index volatility against buying constituent volatility is a short position in implied correlation (§11.7), and its returns are compensation for correlation risk (Driessen, Maenhout and Vilkov, 2009). *Fails when* correlations go to one in a crisis, which is the risk being paid for.

Events. Compare the implied event move (§5.5) with the distribution of the underlying's historical moves on comparable events. *Fails when* the historical sample is short, the company has changed, or the "premium" is really compensation for a genuinely fatter tail (Dubinsky, Johannes, Kaeck and Seeger, 2019).

Cross-sectional volatility. Rank stocks by the spread between trailing realised and implied volatility and trade delta-hedged options accordingly (Goyal and Saretto, 2009). *Fails when* transaction costs in single-stock options — which are wide — eat the spread, which is the standard critique of this family of results.

13.3 Directional options traders

Choosing the structure. The surface tells you what your view costs in each form. In a steep skew, outright puts are expensive and put spreads or put ratios are relatively cheaper; when call skew is flat, overwriting earns little. The right question is never "are options cheap?" but "which part of this surface is expensive relative to my view?"

The expected move. The at-the-money straddle prices the expected absolute move (§2.4), and around events the term-structure decomposition gives the event's own share (§5.5). This is the single most useful number for someone deciding whether an options trade expresses their view: if you think the stock moves 3% on earnings and the options price 7%, buying premium is a bad way to be right.

Probabilities. Use the skew-corrected digital of §7.4, remember it is risk-neutral, and treat any platform's "probability of profit" as a delta unless proven otherwise.

13.4 Risk managers

Scenario grids. Revalue the book over a grid of spot and volatility shocks, with the volatility shock applied under an explicit sticky rule. Short-dated volatility moves more than long-dated in any real shock, so scaling volatility shocks by roughly $\sqrt{30/T}$ across the curve is a common convention. [Practice]

Volatility inputs for risk models. Implied volatility reacts within minutes to a shock while historical estimators take weeks, which makes it attractive for risk measurement; it also embeds the premium of §6.3, which makes it conservative. Both are acceptable as long as the model documentation does not call it an unbiased forecast.

Vega bucketing. Vega is not one number: a book can be flat total vega and badly exposed to the curve steepening or the skew rotating. Bucket by expiry and by moneyness.

Model risk. Marks, reserves and independent price verification all depend on the surface, and the surface depends on the fit. *Fails when* a single vendor surface is used both for marking and for validating the marks.

13.5 Allocators and macro investors

A risk gauge. The VIX is the market's most-watched measure of equity risk appetite — the "investor fear gauge" (Whaley, 2000) — and cross-asset analogues exist for rates, currencies and commodities. *Fails when* it is read as a forecast of direction: a high VIX says options are expensive and recent moves have been large, not that the market will fall.

A regime variable. The level and especially the term-structure slope work well as conditioning variables: an inverted volatility curve is one of the cleaner markers of market stress. Market Regimes and Hidden Markov Models covers how to use such a variable without fooling yourself.

Volatility targeting. Scaling exposure by inverse volatility can use implied rather than realised volatility as the scaler, trading the premium's bias for faster reaction. [Practice] The evidence on volatility-managed portfolios is built on realised measures (Moreira and Muir, 2017), so an implied-volatility variant needs its own testing rather than inheriting those results.

Expected returns. The SVIX bound of Martin (2017) turns option prices into a lower bound on the equity premium, which spiked above 20% in late 2008 and averages around 5%. It is the most direct use of implied volatility in asset allocation.

Hedging cost. How expensive tail protection is depends on skew and the premium, both of which are cheapest when nobody wants them. The structural problem with tail hedging is that its cost is a function of demand for it. [Practice]

13.6 Researchers: implied volatility as a signal

The option surface contains information the underlying's price history does not, and a substantial literature tries to extract it.

Signal	Finding	Status
Variance risk premium	Predicts aggregate equity returns, strongest at a quarterly horizon (Bollerslev, Tauchen and Zhou, 2009)	[Fact] in sample; [Contested] out of sample
Call–put implied volatility spread	Stocks with relatively expensive calls outperform by about 50 bp a week (Cremers and Weinbaum, 2010)	[Fact] in sample
Volatility smirk steepness	Steepest-smirk stocks underperform by about 10.9% a year (Xing, Zhang and Zhao, 2010)	[Fact] in sample
Changes in implied volatility	Large increases in call implied volatility predict high returns; puts the reverse (An, Ang, Bali and Cakici, 2014)	[Fact] in sample
Implied minus trailing realised	Predicts delta-hedged option returns cross-sectionally (Goyal and Saretto, 2009)	[Contested] after costs
Sensitivity to aggregate volatility	Stocks loading positively on volatility innovations earn lower returns (Ang, Hodrick, Xing and Zhang, 2006)	[Fact]
Option-implied political-event premia	Options spanning political events are more expensive (Kelly, Pástor and Veronesi, 2016)	[Fact]

Four cautions apply to all of them. Single-stock option spreads are wide, so paper returns overstate achievable ones. Implied volatilities computed from closing quotes are not tradable at those levels, so a signal built on them can be unimplementable. Stale quotes create look-ahead when a "today" implied volatility reflects a price from hours ago. And this is a heavily mined area, so published effects should be discounted for selection (Dealer Hedging, Appendix A.28, covers the multiple-testing machinery).

For machine-learning features, the useful primitives are the ones this document has been building: level, normalised skew, curvature, term slope, the variance risk premium, and the daily changes in each. Normalise each by its own history, compute everything point-in-time, and never feed a model a raw implied volatility whose units drift with the volatility regime.

13.7 Outside trading

Accounting. Companies expensing employee stock options must estimate expected volatility, and the SEC's guidance explicitly allows a company with actively traded options to place greater — or even exclusive — reliance on implied volatility (SEC Staff Accounting Bulletin 107, 2005). This is implied volatility used as a forecast in a legal document, premium and all.

Central banks and policy analysis. Option-implied densities are a standard tool for reading market expectations around policy decisions: the Bank of England published implied probability density functions for the FTSE 100 and short sterling until 2021, and the Federal Reserve Bank of Minneapolis publishes market-based probabilities of large moves. Both institutions are careful to describe these as market-implied rather than actual probabilities — the $\mathbb{Q}$-versus-$\mathbb{P}$ distinction of §6.1, taken seriously in public.

Corporate and credit applications. Equity implied volatility feeds structural credit models, where it proxies for asset volatility, and equity option prices around announced deals encode the market's view of completion risk. [Practice]

§13 Key takeaways

1. Market makers use the surface as the pricing engine, the marking basis and the hedge-ratio generator; its staleness is their main operational risk.
2. For a volatility trader, the edge must exceed the premium plus hedging error plus costs — several volatility points, not a fraction of one.

3. Selling variance is a sound business with a brutal distribution; size it by stress loss rather than by return volatility.
4. Directional traders should ask which part of the surface is expensive relative to their view, not whether options are cheap.
5. Risk managers get a fast-reacting, conservative input, provided nobody calls it unbiased; bucket vega by expiry and moneyness.
6. Allocators use the level as a risk gauge, the slope as a regime variable, and the SVIX bound as a direct read on expected returns.
7. Option-implied signals predict returns in sample across several independent constructions; costs, stale quotes and selection are the standing objections.
8. Outside trading, implied volatility appears in accounting rules and central-bank market monitoring, in both cases as a forecast that carries a premium.

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14. Failure modes

Everything that goes wrong with implied volatility in practice falls into a short list. This is that list, in the order in which the errors do damage.

Failure	Mechanism	Symptom	Fix	§
Wrong forward	Dividends, borrow or rates assumed rather than implied	Call and put volatilities disagree; a fake skew	Imply the forward from put–call parity	§9.4
Non-synchronous prices	Option quote and underlying from different instants	Spurious skew and noisy time series	Use the implied forward; check timestamps	§9.4
Wrong clock	Calendar versus trading versus variance time	Intraday and weekend volatilities wrong by factors, not percents	State the convention; use a variance clock	§9.4
American exercise	European formula on American prices	In-the-money options show inflated volatility	Use out-of-the-money options or a tree	§9.4
Inverting noise	Vega small in wings and near expiry	Wild wing volatilities; unstable fits	Weight by vega; drop tick-priced options	§9.3
Polynomial smile fits	Quadratic wings violate Lee's bound	Negative densities when extrapolated	Fit in total variance with linear wings (SVI)	§7.5, §10.4
Interpolating volatility in time	Not linear in the additive quantity	Wobbling or negative forward variance	Interpolate total variance	§10.5
Unchecked arbitrage	Fitted surface violates butterfly or calendar	Negative local variance; nonsense exotics	Run the §10.6 checks daily	§10.6
Confusing index with at-the-money	VIX is a variance-swap rate, not an ATM volatility	Apparent "volatility spikes" that are skew moves	Compare like with like	§5.3
$N(d_2)$ as a probability	Ignores skew; wrong measure	Overstated downside probabilities	Use the skew-corrected digital; label it risk-neutral	§7.4
Treating implied as a forecast	Ignores the variance risk premium	Persistent "options are expensive" conclusions	Debias by regression; compare in variance units	§6.3, §6.5
Horizon mismatch	30-day implied against 20-day realised, or trailing instead of forward	Spurious forecast failure	Match horizons exactly; align forward	§11.8, §12.5
Rank versus percentile	One spike sets the range	"Volatility is at 11% of its range" when it is mid-pack	Prefer percentile; show both	§11.8
Unnormalised skew series	Skew scales with level and maturity	A skew time series that is really a volatility series	Normalise by level and $\sqrt{T}$	§11.3
Wrong sticky assumption	Delta depends on surface dynamics	Systematic hedging P&L leakage	Estimate the skew-stickiness ratio	§8.5
Stale surface in a fast market	Fitted yesterday, quoted today	Adverse selection on every quote	Refit intraday; widen when uncertain	§13.1

Three of these deserve more than a table row.

The wrong forward is the most damaging error in the field, because it is invisible. Everything downstream still computes: you get a surface, it has a skew, the skew has a time series, and the time series correlates with things. The tell is the call–put volatility gap at the same strike, which a 0.5% forward error blows out to 4.4 volatility points (§9.4). Check it before anything else, every day, per expiry.

Comparing implied with realised volatility goes wrong in four independent ways at once, and most published "implied volatility is useless" claims trip on at least one: the horizon must match, the alignment must be forward, the comparison should be in

variance units, and the expected gap is not zero but the premium. For a single option, the theoretically correct comparison is against dollar-gamma-weighted realised variance (§4.5), which almost nobody computes and which matters most exactly when the price has wandered far from the strike.

Small arbitrage in a fitted surface becomes large nonsense downstream. A butterfly violation too small to see on a plot is a negative probability, and it becomes a negative local variance (§8.3), an exotic price that is off by a factor, or a VIX-style integral that is quietly wrong. Differentiation amplifies; always check the conditions rather than the picture.

§14 Key takeaways

1. The forward is the first thing to get right and the first thing to check; the call–put volatility gap is the diagnostic.
2. Time conventions are not a detail: the same intraday price implies volatilities that differ by a factor of 2.3 across clocks.
3. Wing implied volatilities are mostly noise; weight by vega and discard tick-priced options.
4. Fit in total variance with linear wings and interpolate total variance in time; anything else manufactures arbitrage.
5. Compare like with like: a variance-swap index is not an at-the-money volatility, and implied is not realised.
6. Every probability from a surface is risk-neutral, and every tail is extrapolated.
7. Check arbitrage conditions numerically rather than visually, because the downstream quantities differentiate the surface.

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15. Synthesis

15.1 The framework on one page

One number, defined by inverting a formula nobody believes, turns out to be three things at once.

A price. Implied volatility is an option's price in the unit where a lognormal, constant-volatility world is flat. Everything the ruler cannot express — fat tails, asymmetry, volatility that moves — shows up as *shape* in the surface rather than as error in the number. This is why the market kept a model it knows to be false: as a coordinate system it is exact, and as a coordinate system it is all anyone needs.

A rate. A delta-hedged option earns half its dollar gamma times realised variance minus implied variance. Implied volatility is therefore the break-even variance rate of the hedge, and — taking expectations under the pricing measure — a dollar-gamma-weighted average of the variance the market prices along the paths the option cares about:

$$\sigma_i^2 = \frac{\mathbb{E}^{\mathbb{Q}}\left[\int_0^T \Gamma_t S_t^2 \sigma_t^2 dt\right]}{\mathbb{E}^{\mathbb{Q}}\left[\int_0^T \Gamma_t S_t^2 dt\right]}.$$

Flatten the weights with a $1/K^2$ strip and you get the variance swap and the VIX. Slice the integral in time and you get forward variance and event moves. Change the measure and you get a forecast, with the variance risk premium as the difference.

A shape. Across strikes, prices *are* the risk-neutral distribution: the second strike-derivative of the call price is the density. The smile is that distribution written in lognormal units, so level is width, slope is skewness, curvature is tails.

The two are one object. Dupire's formula divides the time-direction quantity by the strike-direction quantity and returns local variance, and the conditions for a surface to be arbitrage-free are exactly that this ratio is well behaved:

$$\sigma_{\text{loc}}^2 = \frac{\partial w/\partial T}{g(k)}, \qquad \underbrace{\partial w/\partial T \geq 0}_{\text{no calendar arbitrage}}, \qquad \underbrace{g(k) > 0}_{\text{no butterfly arbitrage}} .$$

15.2 Equivalences

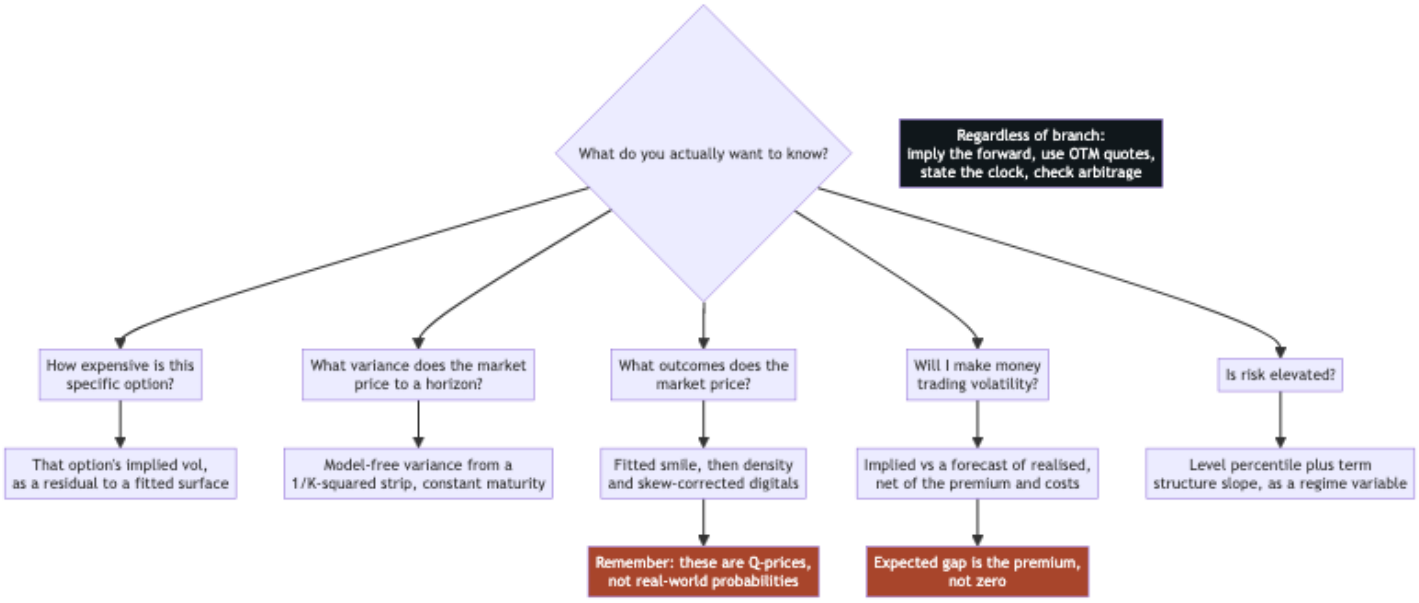
The highest-value half-page in a document like this is usually the list of things that look different and are the same, and things that share a name and are not.

Claim	Status
Option price $\leftrightarrow$ implied volatility	**Exact** one-to-one inside the no-arbitrage bounds (vega > 0)
Call implied volatility $=$ put implied volatility at the same strike	**Exact** for European options with the right forward; deviations diagnose inputs
Theta $= -\frac{1}{2}\sigma_i^2 S^2 \Gamma$	**Exact** under Black–Scholes with zero rates
Implied variance $=$ dollar-gamma-weighted average of priced variance	**Exact** for continuous paths (§4.3)
$1/K^2$ strip $=$ log contract $=$ variance swap	**Exact** for continuous paths; jumps break it at third order
Variance-swap rate $=$ normal-weighted average of implied variance over $z = -d_2$	**Exact** under the same conditions (verified numerically to three decimals, §5.3)
VIX^2 $=$ 30-day variance-swap rate	**Approximate**: discretisation, truncation, jumps
$\partial^2 C / \partial K^2 = D f_T(K)$	**Exact**
Flat smile $\leftrightarrow$ lognormal distribution	**Exact**
No calendar arbitrage $\leftrightarrow$ total variance non-decreasing in expiry	**Exact** at fixed forward-moneyness
No butterfly arbitrage $\leftrightarrow$ $g(k) > 0$ $\leftrightarrow$ positive density	**Exact**
ATM straddle $=$ expected absolute move	**Approximate**: $\sqrt{2/\pi}$ scaling, near-normal returns, and under $\mathbb{Q}$
ATM implied volatility $\approx$ volatility-swap rate	**Approximate**, and better than treating it as a variance-swap rate
Implied skew $=$ half the local skew	**Asymptotic**, short maturities, near the money
Implied volatility $=$ harmonic mean of local volatility from spot to strike	**Asymptotic** (short maturity)
Sticky strike / sticky delta / local volatility $\leftrightarrow$ skew-stickiness ratio 1 / 0 / 2	**Definitional**; real markets sit between 1 and 2

And the traps — same name, different object:

Name used for both	The two things
“Implied volatility”	One option’s inverted price; a VIX-style variance-swap rate; an at-the-money level; a whole surface
“Variance (or volatility) risk premium”	Implied minus expected realised in **volatility** points; the same in **variance** units (sign conventions also differ)
“Forward volatility”	The rate implied between two expiries; the implied volatility of a forward-starting option (different, §11.6)
“At-the-money”	Spot, forward, delta-neutral straddle, 50-delta, nearest listed strike (§11.1)
“IV rank” versus “IV percentile”	Range position versus frequency below (differed by 45 points in §11.8’s example)
“Probability of finishing in the money”	$N(d_2)$; the skew-corrected digital; delta; and all three under $\mathbb{Q}$, not $\mathbb{P}$
“Skew”	The slope of the smile; risk-neutral skewness; a risk reversal; the Cboe SKEW index

15.3 A decision tree



15.4 A staged build, with gates

If the goal is a working implied-volatility pipeline, build it in this order and do not pass a stage until its gate is clean. The early stages are unglamorous infrastructure, and skipping them is the usual cause of failure.

Stage	What you build	Gate
0. Data	Quotes with bid, ask, size, timestamp; underlying prices; rates; expiry calendars	Timestamps align; expiry dates and settlement times are right
1. Forward	Implied forward and discount per expiry from put–call parity	Call and put implied volatilities agree at each strike within the quote width
2. Inversion	Implied volatilities for out-of-the-money options, with bid/ask volatilities	Round-trip: reprice from the volatility and recover the input price
3. Smile fit	One arbitrage-free slice per expiry in total variance	Fit inside the bid–ask on liquid strikes; $g(k) > 0$ everywhere
4. Surface	Expiries joined by total-variance interpolation; constant-maturity series	Calendar condition holds; the constant-maturity series has no sawtooth
5. Derived	ATM, skew, term slope, model-free variance, densities, event moves	Your VIX reproduces the published VIX to a few hundredths
6. Research	History, premia, signals	No look-ahead; horizons matched; results survive costs

15.5 Ten things I would tell someone starting today

1. **Get the forward right before anything else.** Imply it from put–call parity, check the call–put volatility gap daily, and treat a disagreement as a data bug rather than a trading signal.
2. **Work in total variance and log-moneyness.** Nearly every formula gets simpler, the no-arbitrage conditions become checkable, and interpolation stops manufacturing structure.
3. **Say which volatility you mean.** A VIX-style index, an at-the-money level and a 25-delta put volatility are three different numbers, and most confused conversations about "volatility" are two people holding different ones.
4. **Expect implied to exceed realised.** Four volatility points on the S&P 500 over three decades. A strategy whose thesis is "options are expensive" has discovered the premium, not an edge.
5. **Never read probabilities off $N(d_2)$ at the strike's own volatility.** Use the skew-corrected digital, and remember it is still risk-neutral.
6. **Throw away the wings and short-dated quotes that vega says are noise.** Weight fits by vega; a five-cent error is hundreds of volatility points in a seven-day wing.
7. **Check arbitrage numerically, not visually.** Butterfly and calendar violations too small to see become negative densities and nonsense local variance.
8. **Decide your sticky rule explicitly.** Your delta depends on it, and the empirical answer for index options is somewhere between sticky strike and local volatility.
9. **Match horizons and align forward when comparing implied with realised.** Most claims that implied volatility fails as a forecast are horizon or alignment errors.
10. **Reproduce the published VIX as a test of your whole pipeline.** It exercises the forward, the filters, the clock and the interpolation simultaneously, and it is the only end-to-end check with a published answer.

15.6 What is known, and what is not

Known, and not seriously disputed. Implied volatility is a price, not a forecast. The inversion is well defined and its conditioning is understood. The variance-swap replication is exact for continuous paths and is the basis of every volatility index. Option prices across strikes determine the risk-neutral distribution exactly. Equity index skew has been persistently negative since 1987. Implied variance exceeds subsequently realised variance on average for equity indices, by a margin large enough to support an industry, and that premium is compensation for a genuinely bad payoff distribution. Surfaces move mostly in level, and level moves are negatively correlated with the underlying.

Contested, with live disagreement. How much of the index skew is crash risk in the physical distribution versus a price charged by constrained intermediaries. Whether the at-the-money skew follows a power law all the way to zero maturity, and therefore whether "rough volatility" is the right description of the short end. Whether the option-implied predictive signals survive costs and out-of-sample testing. Whether the real-world distribution can be recovered from prices at all. The right skew-stickiness ratio, which appears to be regime-dependent.

Not known, and I would not pretend otherwise. What the variance risk premium *should* be, in the sense of a model that predicts its level rather than measuring it. How much of the short end's behaviour is genuine information about jump risk and how much is the market microstructure of options that expire in hours. Whether the surface's dynamics are stable enough for any fitted sticky rule to hold up through the next regime change. If I had to bet on where the next decade's useful results come from, it would be the interaction between implied volatility and dealer positioning at very short horizons — the subject the Dealer Hedging note takes up from the other side.

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16. References

Grouped by kind. Where a free copy exists it is the link; paywalled-only entries are marked. Practitioner sources from parties with a commercial interest in the conclusion are flagged.

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- **Kelly, B., Pástor, Ľ. & Veronesi, P. (2016).** "The Price of Political Uncertainty: Theory and Evidence from the Option Market." *Journal of Finance* 71(5), 2417–2480. — Options spanning political events are more expensive (§5.5, §13.6).
- **Johnson, T. L. (2017).** "Risk Premia and the VIX Term Structure." *Journal of Financial and Quantitative Analysis* 52(6), 2461–2490. — The slope of the curve prices variance risk rather than forecasting the level (§11.3, §13.2).
- **Moreira, A. & Muir, T. (2017).** "Volatility-Managed Portfolios." *Journal of Finance* 72(4), 1611–1644. — Scaling exposure by volatility; the evidence base for volatility targeting (§13.5).
- **Dubinsky, A., Johannes, M., Kaeck, A. & Seeger, N. J. (2019).** "Option Pricing of Earnings Announcement Risks." *Review of Financial Studies* 32(2), 646–687. — Separating event variance from diffusive variance, and what the implied move is worth (§5.5, §13.2).
- **US Securities and Exchange Commission (2005).** "Staff Accounting Bulletin No. 107." — Guidance permitting exclusive reliance on implied volatility when estimating expected volatility for option expensing (§13.7).
- **Bank of England.** "Option-Implied Probability Density Functions." — Weekly implied densities for the FTSE 100 and short sterling, discontinued in 2021; the documentation is a good short guide to the method (§11.5, §13.7).
- **Federal Reserve Bank of Minneapolis.** "Market-Based Probabilities." — Option-implied probabilities of large moves, published and maintained (§11.5, §13.7).

16.9 Books and practitioner sources

- **Taleb, N. N. (1997).** *Dynamic Hedging: Managing Vanilla and Exotic Options.* Wiley. — What hedging an option book actually involves, including the parts models omit.
- **Rebonato, R. (1999).** "Volatility and Correlation in the Pricing of Equity, FX and Interest-Rate Options." Wiley; 2nd edition (2004) retitled *Volatility and Correlation: The Perfect Hedger and the Fox*. — Source of the description of implied volatility quoted in §1.2, and a careful treatment of what the number does and does not mean.
- **Gatheral, J. (2006).** "The Volatility Surface: A Practitioner's Guide." Wiley. — The single best book on this subject: local volatility, the variance-swap formula of §5.3, and the surface's dynamics.

- **Sinclair, E. (2013).** “Volatility Trading.” 2nd ed., Wiley. — The trading side: forecasting, position sizing, and evaluating volatility trades.
- **Natenberg, S. (2015).** “Option Volatility and Pricing.” 2nd ed., McGraw-Hill. — The book trading firms hand new options traders.
- **Hull, J. C. (2021).** “Options, Futures, and Other Derivatives.” 11th ed., Pearson. — The standard textbook for contract mechanics and pricing.

16.10 If you only read eight things

In order:

1. **Demeterfi, Derman, Kamal and Zou (1999)** — the replicating strip, explained so clearly that the VIX methodology cites it.
2. **Lee (2005)** — what implied volatility means, surveyed by someone who proved several of the results.
3. **Breeden and Litzenberger (1978)** — four pages that turn option prices into a probability distribution.
4. **Gatheral (2006)** — the book to own if you work with surfaces.
5. **Canina and Figlewski (1993), then Christensen and Prabhala (1998)** — a textbook case of an empirical debate resolved by better statistics.
6. **Carr and Wu (2009)** — the variance risk premium, measured properly.
7. **Gatheral and Jacquier (2014)** — how to fit a surface that cannot be arbitraged.
8. **Derman (1999)** — six pages on how the surface moves, which is the part that decides your hedge.

Appendix A. Concepts and prerequisites

Every concept the main text uses without fully explaining, ordered so that each entry leans only on the ones before it. Each entry gives the idea first, in plain language; then the formal statement; then where the main text relies on it; then where to read further. A reader who stops after “The idea” should still come away with something usable. A.1–A.5 are the probability behind the main text, A.6–A.11 the pricing machinery, A.12–A.16 models of the surface, A.17–A.25 statistics and econometrics, and A.26–A.32 numerics and market practice.

Concept	Where the main text leans on it	Entry
Brownian motion, and why variance grows with time	§2.1, §4.3, §4.4	A.1
Quadratic variation, realised variance, and jumps	§5.1, §5.6	A.2
Itô’s lemma	§4.1, §4.3	A.3
Martingales, and why a hedging gain has zero expected value	§4.3, §5.1	A.4
The Brownian bridge	§4.4, §8.2	A.5
State prices, the pricing measure, and the stochastic discount factor	§6.1, §7.2, §7.4, §12.4	A.6
Convexity, Jensen’s inequality, and convexity adjustments	§4.3, §5.3, §6.1	A.7
European and American exercise	§9.2, §9.4, §10.2	A.8
Static replication, and the spanning formula	§5.1, §7.3, §11.4	A.9
SVIX, and option-implied expected returns	§11.4, §13.5	A.10
Second-order Greeks: vomma and vanna	§4.3, §8.4	A.11
Stochastic volatility models	§4.3, §8.4, §10.4	A.12
Jump-diffusion models	§5.6, §8.4	A.13
Fractional Brownian motion, the Hurst exponent, and rough volatility	§1.5, §8.4, §11.3	A.14
The SVI and SSVI parameterisations	§10.3, §10.4, §10.6	A.15
The minimum-variance delta	§8.5, §13.1	A.16
Skewness, kurtosis, and the Gram–Charlier expansion	§7.3	A.17
History-based volatility forecasts: GARCH and HAR	§6.2, §6.5, §12.6	A.18
Measurement error, attenuation, and instrumental variables	§6.2	A.19
Overlapping observations, and why they inflate significance	§6.2, §12.5	A.20
Loss functions for volatility forecasts	§6.2	A.21
Principal components	§8.6, §12.3	A.22
Beta, and why insurance earns a negative return	§6.4, §13.6	A.23
Index variance, implied correlation, and dispersion	§11.7, §13.2	A.24
The recovery theorem, and why it is contested	§6.5	A.25
Root-finding: bisection, Newton, and higher-order methods	§9.2	A.26
Conditioning, and why numerical derivatives amplify noise	§9.3, §10.6, §11.5, §14	A.27
Cubic splines	§11.5	A.28
Binomial trees for American options	§9.2, §9.4	A.29
Delta conventions, and delta as a strike coordinate	§10.3, §10.4, §11.1, §11.3	A.30
Look-ahead bias, and data snooping	§12.5, §13.6, §15.4	A.31

Concept	Where the main text leans on it	Entry
Structural credit models	§13.7	A.32

A.1 Brownian motion, and why variance grows with time

The idea. Take a random walk with many tiny independent steps and shrink the steps toward zero while speeding up the clock. The limit is Brownian motion: a path that is continuous but jagged at every scale, whose variance grows in proportion to elapsed time. It is the noise source of every model in this document, and the reason option prices depend on total volatility $\sigma\sqrt{\tau}$ rather than on σ and τ separately.

Formally. A standard Brownian motion W_t starts at zero, has independent increments, and has $W_t - W_s \sim N(0, t-s)$ for $s < t$, with continuous paths. Geometric Brownian motion, $dS_t = \mu S_t\,dt + \sigma S_t\,dW_t$, has the solution $S_T = S_0 \exp\big((\mu - \tfrac{1}{2}\sigma^2)T + \sigma W_T\big)$, so $\ln S_T$ is normal and S_T lognormal.

Why it appears here. The $\sqrt{\tau}$ scaling of §2.1; the process $dS_t = \sigma_t S_t\,dW_t$ of §4.3; the bridge of §4.4 (A.5).

Deeper. Stochastic Processes builds it from the ground up; Dealer Hedging, Appendix A.2, connects it to option pricing.

A.2 Quadratic variation, realised variance, and jumps

The idea. Add up the squared changes of a path over many small intervals. For a smooth curve the sum vanishes as the intervals shrink; for a Brownian path it converges to the elapsed time, and for a price path to the integrated variance. This limit is what "realised variance" estimates, and what a variance swap pays. Jumps add their own squared size to it, which is where the variance-swap replication of §5 stops being exact.

Formally. The quadratic variation of X over $[0,T]$ is $[X]_T = \lim \sum_k (X_{t_{k+1}} - X_{t_k})^2$ as the mesh goes to zero. For $d\ln S_t = (\ldots)\,dt + \sigma_t\,dW_t$ plus jumps, $[\ln S]_T = \int_0^T \sigma_t^2\,dt + \sum_{\text{jumps}} (\Delta \ln S)^2$. Realised variance from daily log returns, $\sum_k r_k^2$, estimates it.

Why it appears here. §5.1 (the strip pays integrated variance) and §5.6 (a jump of -20 log points adds 0.04 to quadratic variation but only 0.0375 to the log contract).

Deeper. Carr and Lee (2009); Dealer Hedging, Appendix A.3.

A.3 Itô's lemma

The idea. The ordinary chain rule fails for functions of Brownian motion, because the squared increment $(dW)^2$ is of order dt rather than negligibly small. The correction is one extra second-derivative term, and in option pricing that term is exactly where gamma meets variance: it is the $\frac{1}{2}\Gamma S^2\sigma^2$ that the hedged-P&L identity is built from.

Formally. For $V(S,t)$ with $dS_t = \sigma_t S_t\,dW_t$,

$$dV = \big(\partial_t V + \tfrac{1}{2}\sigma_t^2 S_t^2\,\partial_S^2 V\big)\,dt + \partial_S V\,dS_t.$$

Why it appears here. The first step of the derivation in §4.3, and implicitly the identity of §4.1.

Deeper. Dealer Hedging, Appendix A.3; Stochastic Processes.

A.4 Martingales, and why a hedging gain has zero expected value

The idea. A martingale is a fair game: given everything known today, the expected future value is today's value. Under the pricing measure, discounted prices are martingales, and so is the gain from any reasonable trading strategy in them — you cannot expect to profit from trading a fairly priced asset. That is why, in §4.3, the term $\int \Delta\,dS$ disappears when expectations are taken.

Formally. X is a martingale if $\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s$ for $s \le t$, where $\mathcal{F}_s$ is the information available at s. If S is a $\mathbb{Q}$-martingale and Δ is adapted and suitably integrable, $\mathbb{E}^{\mathbb{Q}}\big[\int_0^T \Delta_t\,dS_t\big] = 0$.

Why it appears here. §4.3, and the replication arguments of §5.1.

Deeper. Dealer Hedging, Appendices A.5–A.6.

A.5 The Brownian bridge

The idea. A Brownian motion forced to finish at a chosen point. Its paths start where they must, end where they must, and wander in between, with the most freedom in the middle of the interval and none at either end. Its mean path is the straight line between the endpoints.

Formally. For $X_t = x_0 + \sigma W_t$ conditioned on $X_T = x_T$,

$$X_t \sim N\left(x_0 + (x_T - x_0)\frac{t}{T},\ \sigma^2\,\frac{t(T-t)}{T}\right).$$

Why it appears here. §4.4. Under Black–Scholes dynamics, the dollar gamma of a strike-K option at log-price x and time t is K times the probability density of travelling from (x, t) to $(\ln K, T)$. Multiply by the density of reaching (x, t) from today, and the product of the two densities is the density of a bridge from $\ln S_0$ to $\ln K$, times a constant. That is why the weights in the figure of §4.4 form a lens from today's price to the strike, and why their total per unit time is constant.

Deeper. Gatheral (2006), in the discussion of implied volatility as an average of local volatility.

A.6 State prices, the pricing measure, and the stochastic discount factor

The idea. A state price is today's cost of receiving one dollar in one particular future state of the world. Rescale all state prices so they sum to one and they look like probabilities: that is the pricing measure $\mathbb{Q}$. States in which a dollar is especially valuable — crashes, recessions — have high state prices relative to how often they occur, so $\mathbb{Q}$ overweights them relative to the real-world measure $\mathbb{P}$. The ratio between the two measures is proportional to the **stochastic discount factor**, the marginal value of a dollar in each state.

Formally. For a payoff X at T, price $= \mathbb{E}^{\mathbb{P}}[mX] = D\,\mathbb{E}^{\mathbb{Q}}[X]$, with $d\mathbb{Q}/d\mathbb{P} = m/\mathbb{E}^{\mathbb{P}}[m]$ (the Radon–Nikodym derivative), where m is high in bad states.

Why it appears here. §6.1 (implied volatility is a $\mathbb{Q}$ quantity), §7.2 (the Breeden–Litzenberger density is a $\mathbb{Q}$ density), §7.4 and §12.4 (option-implied probabilities overstate real-world ones for bad states).

Deeper. Breeden and Litzenberger (1978); Dealer Hedging, Appendix A.6; A.25 for the inverse problem.

A.7 Convexity, Jensen's inequality, and convexity adjustments

The idea. For a convex function, the average of the function exceeds the function of the average; for a concave one, the reverse. The square root is concave, so the average of volatilities is below the square root of the average variance — which is why a volatility swap is worth less than the square root of a variance swap, and why comparisons made in volatility units and in variance units disagree. Convexity also drives the smile: an option's price is convex in volatility away from the money, so uncertainty about volatility raises out-of-the-money prices.

Formally. Jensen: $f(\mathbb{E}[X]) \le \mathbb{E}[f(X)]$ for convex f. For a variance V with mean μ, a second-order expansion of $\sqrt{V}$ about μ gives $\mathbb{E}[\sqrt{V}] \approx \sqrt{\mu} - \mathrm{Var}(V)/(8\mu^{3/2})$.

Why it appears here. §4.3 (the smile from random volatility), §5.3 (volatility swap versus variance swap), §6.1 (premia in volatility versus variance units).

Deeper. Carr and Lee (2009).

A.8 European and American exercise

The idea. A European option can be exercised only at expiry; an American option at any time before. The right to exercise early is worth something when holding the option costs more than exercising it — for a deep in-the-money put, receiving the strike now and earning interest on it; for a call, capturing a dividend. An American option is therefore worth at least as much as its European twin, and the difference (the early-exercise premium) has no closed form.

Formally. $V^{\mathrm{Am}} = \sup_{\theta \le T} \mathbb{E}^{\mathbb{Q}}\big[D(\theta)\,\mathrm{payoff}(S_\theta)\big]$ over stopping times θ (rules for when to exercise that use only information available at the time).

Why it appears here. §9.2 and §9.4 (inverting American prices with a European formula overstates volatility) and §10.2 (why surfaces are built from out-of-the-money options, whose early-exercise premium is small).

Deeper. Hull (2021).

A.9 Static replication, and the spanning formula

The idea. Any payoff that depends only on the final price can be assembled from a bond, a forward and a continuum of out-of-the-money options, where the amount held at each strike is the payoff's curvature there. Options are the market's supply of curvature; everything else is linear. This is the engine behind the variance swap, the risk-neutral moments, and SVIX.

Formally. For twice-differentiable f,

$$f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F f''(K)(K - S_T)^+ \, dK + \int_F^\infty f''(K)(S_T - K)^+ \, dK.$$

For the log contract $f(S) = -\frac{2}{T}\ln(S/F)$, $f''(K) = 2/(TK^2)$: the $1/K^2$ strip of §5.1.

Why it appears here. §5.1, §7.3 (moment contracts) and §11.4 (SVIX's equal weights).

Deeper. Carr and Madan (2001); Demeterfi, Derman, Kamal and Zou (1999).

A.10 SVIX, and option-implied expected returns

The idea. The $1/K^2$ strip of §5.1 prices the variance of the *log* return. Weight every strike equally instead and the strip prices the risk-neutral variance of the *simple* return. Martin showed that under a weak condition on how the market return co-moves with the stochastic discount factor (A.6), that risk-neutral variance, divided by the gross risk-free rate, is a lower bound on the market's expected excess return. Option prices therefore give a real-time floor under the equity premium.

Formally. Applying the spanning formula (A.9) to $f(S) = S^2$, whose second derivative is the constant 2, gives the first expression below for the gross market return R_T and gross risk-free return R_f; the second is Martin's bound.

$$\text{var}^{\mathbb{Q}}(R_T) = \frac{2R_f}{S_0^2}\int_0^\infty \text{OTM}(K)\, dK, \qquad \mathbb{E}[R_T] - R_f \;\geq\; \frac{\text{var}^{\mathbb{Q}}(R_T)}{R_f}.$$

Why it appears here. §11.4 and §13.5.

Deeper. Martin (2017).

A.11 Second-order Greeks: vomma and vanna

The idea. Vega measures sensitivity to volatility; **vomma** (or volga) measures how vega itself changes with volatility — convexity in volatility. It is positive away from the money, so owning out-of-the-money options is a position in the *volatility of volatility*, and a market that prices uncertainty about volatility lifts the wings: that is smile curvature. **Vanna** measures how delta changes with volatility, or equivalently how vega changes with the price. It is what makes a position exposed to the *correlation* between price and volatility, and so it is tied to skew.

Formally. vomma $= \partial^2 V/\partial\sigma^2 = \nu\, d_1 d_2/\sigma$, which is negative only in the narrow band where d_1 and d_2 have opposite signs, around the forward; vanna $= \partial^2 V/\partial S\,\partial\sigma = -\varphi(d_1)\, d_2/\sigma$ (zero rates and dividends).

Why it appears here. §4.3 (random volatility produces a smile through vomma) and §8.4 (volatility of volatility sets curvature; spot–volatility correlation sets skew).

Deeper. Dealer Hedging, section 3.8, for vanna; Bergomi (2016) for how these sensitivities organise smile dynamics.

A.12 Stochastic volatility models

The idea. Let volatility follow its own random process, correlated with the price. Two effects follow directly. Uncertainty about how much variance will arrive makes the distribution of returns a mixture of normals with different widths — fat tails, hence smile curvature. Correlation between price and volatility makes large down-moves more likely to happen in high-volatility states — asymmetry, hence skew. The models differ mainly in how volatility mean-reverts and how its own volatility scales.

Formally. Heston: $dS_t = \sqrt{v_t}\, S_t\, dW_t$ and $dv_t = \kappa(\theta - v_t)\, dt + \xi\sqrt{v_t}\, dZ_t$ with $\text{corr}(dW, dZ) = \rho$: mean reversion speed κ, long-run variance θ, volatility of volatility ξ. Hull–White is the case in which volatility is independent of the price ($\rho = 0$), where the option price is the Black–Scholes price averaged over the distribution of realised average variance.

SABR models the forward directly, $dF_t = \alpha_t F_t^{\beta}\, dW_t$ with $d\alpha_t = \xi\, \alpha_t\, dZ_t$ (the original paper writes ν for ξ), and comes with a closed-form approximate implied volatility.

Why it appears here. §4.3 (the special cases of the master equation), §8.4 (what generates a smile) and §10.4 (SABR as a fitting tool).

Deeper. Hull and White (1987); Heston (1993); Hagan, Kumar, Lesniewski and Woodward (2002); Bergomi (2016).

A.13 Jump-diffusion models

The idea. Add occasional discontinuous jumps to a diffusion. Over a very short horizon a diffusion almost cannot travel far, while a jump can travel any distance at once, so jumps make short-dated out-of-the-money options far more valuable than any diffusion allows — a steep short-dated smile that flattens as maturity grows and the diffusion catches up. Jumps with a negative average size make that smile a skew.

Formally. Merton: $dS_t/S_{t^-} = (r - \lambda\bar{k})\, dt + \sigma\, dW_t + (J-1)\, dN_t$, where N is a Poisson process with intensity λ, ln J is normal, and $\bar{k} = \mathbb{E}[J-1]$ keeps the discounted price a martingale. Under a pure diffusion, an out-of-the-money option's price vanishes faster than any power of T as expiry approaches; with jumps it vanishes only in proportion to T — the distinction that the test of Carr and Wu (2003) exploits.

Why it appears here. §5.6 (jumps break the variance-swap replication) and §8.4.

Deeper. Merton (1976); Bates (2000); Carr and Wu (2003).

A.14 Fractional Brownian motion, the Hurst exponent, and rough volatility

The idea. Brownian motion has independent increments. Fractional Brownian motion relaxes that: with Hurst exponent $H > \frac{1}{2}$ increments are positively correlated and paths look smoother and trendier; with $H < \frac{1}{2}$ they are negatively correlated and paths are rougher than Brownian motion. "Rough volatility" is the finding that the logarithm of volatility behaves like fractional Brownian motion with H around 0.1, and its most visible consequence for options is an at-the-money skew that grows without bound as maturity shrinks, roughly as $T^{H-1/2}$.

Formally. B^H is a Gaussian process with $B_0^H = 0$ and $\mathbb{E}\big[(B_t^H - B_s^H)^2\big] = |t-s|^{2H}$; $H = \frac{1}{2}$ recovers Brownian motion.

Why it appears here. §1.5, §8.4 and §11.3 (the power-law term structure of skew, and the dispute over its shortest end).

Deeper. Gatheral, Jaisson and Rosenbaum (2018); Guyon and El Amrani (2022) for the critique.

A.15 The SVI and SSVI parameterisations

The idea. SVI ("stochastic volatility inspired") describes one expiry's total implied variance as a hyperbola in log-moneyness: two straight wings joined smoothly around the money. Straight wings are what stochastic volatility models produce far from the money, and what Lee's bound (§7.5) permits, and the five parameters each have a visual meaning. SSVI ("surface SVI") ties the expiries together: every slice is determined by its own at-the-money total variance θ and one shared shape function $\phi(\theta)$. That is exactly the "level, term structure and one shape" structure that the rescaled panel of §10.3 displays.

Formally. Raw SVI: $w(k) = a + b\big[\rho(k-m) + \sqrt{(k-m)^2 + s^2}\big]$, with asymptotic slopes $b(1-\rho)$ on the left and $b(1+\rho)$ on the right. SSVI: $w(k,\theta) = \frac{\theta}{2}\big[1 + \rho\phi(\theta)k + \sqrt{(\phi(\theta)k + \rho)^2 + 1 - \rho^2}\,\big]$. Gatheral and Jacquier give simple sufficient conditions on ρ and ϕ under which an SSVI surface is free of static arbitrage; the surface in §10.3 satisfies them, and the numerical checks of §10.6 confirm it.

Why it appears here. §10.3, §10.4 and §10.6.

Deeper. Gatheral and Jacquier (2014); Gatheral (2006).

A.16 The minimum-variance delta

The idea. The Black–Scholes delta says how much an option moves when the underlying moves *and nothing else changes*. But implied volatility does change when the underlying moves — for equity indices, systematically falling on rallies and rising on sell-offs. The minimum-variance delta is the hedge ratio that minimises the variance of the hedged position once that typical co-movement is included.

Formally. $\Delta_{\text{MV}} = \Delta^{\text{BS}} + \nu\,\mathbb{E}[d\sigma_i \mid dS]/dS$, the expectation taken at a fixed strike. With a negative skew and a skew-stickiness ratio above one (§8.5), fixed-strike implied volatility tends to fall as the price rises, so $\Delta_{\text{MV}} < \Delta^{\text{BS}}$ for calls.

Why it appears here. §8.5 and §13.1.

Deeper. Hull and White (2017).

A.17 Skewness, kurtosis, and the Gram–Charlier expansion

The idea. Skewness is the standardised third moment: a measure of asymmetry, negative when the left tail is longer. Excess kurtosis is the standardised fourth moment minus 3: a measure of how much probability sits in the tails (and, to compensate, in the centre) relative to a normal distribution. A Gram–Charlier expansion approximates a density that is *close to* normal by multiplying the normal density by a polynomial correction whose coefficients are exactly these two numbers.

Formally. For a standardised variable z, $f(z) \approx \varphi(z)\big[1 + \frac{\gamma_1}{6}\text{He}_3(z) + \frac{\gamma_2}{24}\text{He}_4(z)\big]$ with Hermite polynomials $\text{He}_3(z) = z^3 - 3z$ and $\text{He}_4(z) = z^4 - 6z^2 + 3$. The approximation can go negative when γ_1 or γ_2 is large, which is why §7.3 treats the smile formula derived from it as a qualitative map only.

Why it appears here. §7.3.

Deeper. Corrado and Su (1996); Bakshi, Kapadia and Madan (2003) for the model-free moments.

A.18 History-based volatility forecasts: GARCH and HAR

The idea. Volatility clusters: calm follows calm and turbulence follows turbulence. GARCH turns that into a recursion in which today's variance is a long-run level plus a reaction to yesterday's squared return plus a large share of yesterday's variance. HAR regresses realised variance on its own averages over the last day, week and month — a crude but hard-to-beat approximation to long memory. These are the history-based forecasts that implied volatility is compared against.

Formally. GARCH(1,1): $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta\sigma_{t-1}^2$, with persistence $\alpha + \beta < 1$ and long-run variance $\omega/(1-\alpha-\beta)$. HAR: $RV_{t+1} = c + \beta_d RV_t + \beta_w RV_t^{(5)} + \beta_m RV_t^{(22)} + \varepsilon$, where the superscripts denote trailing averages over 5 and 22 days.

Why it appears here. §6.2 and §6.5; the simulated returns behind the figure of §12.6 are a GARCH(1,1).

Deeper. Poon and Granger (2003); Dealer Hedging, Appendices A.11 and A.24.

A.19 Measurement error, attenuation, and instrumental variables

The idea. When a regressor is measured with noise, the regression slope is biased toward zero: the noise looks like variation in the regressor that the dependent variable fails to respond to. Implied volatility is measured with noise (bid–ask bounce, stale quotes, the input errors of §9), so a regression of realised on implied volatility understates how informative implied volatility is. The fix is to replace the noisy regressor with its projection on an **instrument** — a variable correlated with the true regressor but not with the noise, such as lagged implied volatility.

Formally. If $x = x^* + u$ with u independent noise, then the ordinary least squares slope converges to $\beta\ \text{Var}(x^*)/\big(\text{Var}(x^*) + \text{Var}(u)\big)$.

Why it appears here. §6.2 (the resolution of the forecasting debate).

Deeper. Christensen and Prabhala (1998).

A.20 Overlapping observations, and why they inflate significance

The idea. Regress 30-day realised volatility on 30-day implied volatility sampled every day, and consecutive observations share 29 of their 30 days. The regression errors are then heavily autocorrelated, the effective number of independent observations is about one thirtieth of the nominal count, and naive standard errors are far too small. The remedies are to sample without overlap (monthly, for a monthly horizon) or to use standard errors that account for autocorrelation, with a lag window at least as long as the overlap.

Why it appears here. §6.2, §12.5.

Deeper. Christensen and Prabhala (1998) for a non-overlapping design.

A.21 Loss functions for volatility forecasts

The idea. True variance is never observed; it is estimated by realised variance, which is noisy. Some loss functions still rank competing forecasts correctly on average when scored against a noisy but unbiased proxy — squared error in variance, and QLIKE — and others do not. QLIKE also penalises under-prediction more heavily than over-prediction, which matches how risk managers are hurt by forecast errors.

Formally. For proxy RV and forecast h, $\text{QLIKE}(RV, h) = RV/h - \ln(RV/h) - 1$, which is zero when $h = RV$.

Why it appears here. §6.2.

Deeper. Patton (2011); Dealer Hedging, Appendix A.25.

A.22 Principal components

The idea. Given many correlated series — the implied volatilities at every point of a surface — find the few directions along which they tend to move together. The first direction explains the most variance, the second the most of what remains, and so on. For implied-volatility surfaces the leading directions are recognisable: the whole surface moving up and down (level), the short end moving against the long end or the wings against the middle (slope and skew), and bending (curvature).

Formally. The eigenvectors of the covariance matrix of daily surface changes, ordered by eigenvalue; each eigenvalue is the variance explained by its component. For a surface treated as a continuous function, the same decomposition is called Karhunen–Loève.

Why it appears here. §8.6 and §12.3.

Deeper. Cont and da Fonseca (2002).

A.23 Beta, and why insurance earns a negative return

The idea. An asset's beta is its sensitivity to the market: the covariance of its return with the market's, divided by the market's variance. Assets that pay off when the market falls have negative beta and are valuable as hedges, so investors accept a low or even negative expected return for holding them — the economics of insurance. Long volatility is such an asset, which is the simplest explanation of why implied variance exceeds expected realised variance.

Formally. $\beta_i = \text{Cov}(R_i, R_m)/\,\text{Var}(R_m)$; in the capital asset pricing model, expected excess return is proportional to β_i.

Why it appears here. §6.4 and §13.6.

Deeper. Ang, Hodrick, Xing and Zhang (2006); Bakshi and Kapadia (2003).

A.24 Index variance, implied correlation, and dispersion

The idea. An index is a portfolio, so its variance is the weighted sum of the covariances of every pair of constituents. Each constituent's volatility is known from its own options and the index's volatility from index options; the only unknown left is how correlated the constituents are, and backing it out gives implied correlation. A dispersion trade sells index volatility and buys constituent volatility: it profits when realised correlation comes in below implied, and loses when correlations jump toward one in a crisis, which is the risk its premium pays for.

Formally. $\sigma^2_{\text{idx}} = \sum_j \omega_j^2 \sigma_j^2 + \sum_{j \neq l} \omega_j \omega_l \rho_{jl} \sigma_j \sigma_l$; setting every $\rho_{jl} = \rho$ gives the formula of §11.7. With N equal weights and a common constituent volatility $\bar{\sigma}$, implied correlation is close to $(\sigma_{\text{idx}}/\bar{\sigma})^2$ for large N: $(18/30)^2 = 0.36$ against the exact 0.35 of §11.7.

Why it appears here. §11.7 and §13.2.

Deeper. Driessen, Maenhout and Vilkov (2009); Dealer Hedging, Appendix A.22.

A.25 The recovery theorem, and why it is contested

The idea. Option prices reveal $\mathbb{Q}$, which is the real-world measure multiplied by a risk adjustment; from prices alone only the product is visible. Ross showed that under two strong assumptions — the state of the economy follows a finite Markov chain, and the risk adjustment takes a specific transition-independent form — linear algebra (the Perron–Frobenius theorem) separates the two. The critique is that the assumption itself does the separating: in general the recovered measure differs from the true one by an unidentified long-run risk adjustment.

Why it appears here. §6.5.

Deeper. Ross (2015); Borovička, Hansen and Scheinkman (2016).

A.26 Root-finding: bisection, Newton, and higher-order methods

The idea. To solve $f(\sigma) = 0$, bisection keeps a bracket known to contain the root and halves it each step: slow but unconditionally safe. Newton's method follows the tangent line to where it crosses zero: extremely fast near the root, because each step roughly doubles the number of correct digits, but liable to overshoot badly where the function is nearly flat or changes curvature. Householder methods use higher derivatives to converge faster still; order four roughly quadruples the correct digits per step.

Formally. Newton: $\sigma_{n+1} = \sigma_n - f(\sigma_n)/f'(\sigma_n)$, with error $e_{n+1} \approx \frac{f''}{2f'}\, e_n^2$ near the root (quadratic convergence). Bisection's error halves each step, so reaching machine precision (about 2×10^{-16}) from a bracket of width 10 takes about 55 steps.

Why it appears here. §9.2 (why Newton fails from a low starting volatility in the wings, and why the Manaster–Koehler start fixes it).

Deeper. Manaster and Koehler (1982); Jäckel (2015).

A.27 Conditioning, and why numerical derivatives amplify noise

The idea. A problem is ill-conditioned when a small error in its input produces a large error in its output. Inverting a price into a volatility is ill-conditioned wherever vega is small (§9.3). Differentiating noisy data is ill-conditioned everywhere: a finite difference divides the noise by the step size, and a second difference divides it by the square of the step. That is why densities must be taken from a fitted smile rather than from raw prices, and why tiny arbitrage violations in a fit become large errors in anything computed from its derivatives.

Formally. $\Delta\sigma_i \approx \Delta V/\nu$. For a second difference with step h and independent price noise of standard deviation ϵ, the error in the estimated density is of order ϵ/h^2.

Why it appears here. §9.3, §10.6, §11.5 and §14.

Deeper. Hentschel (2003); Malz (2014).

A.28 Cubic splines

The idea. A cubic spline joins a sequence of points with cubic pieces whose values, slopes and curvatures agree where they meet, giving a smooth curve through the data. A *clamped* spline also fixes the slopes at the two ends, which is how a density recipe keeps the curve from swinging wildly beyond the last quoted strike.

The alternative is to treat density recovery as a regularised inverse problem: choose the smoothest density that reprices the quoted options to within their bid–ask spreads.

Why it appears here. §11.5 (interpolating implied volatility in delta space before differentiating).

Deeper. Malz (2014); Jackwerth and Rubinstein (1996) for the regularised alternative.

A.29 Binomial trees for American options

The idea. Chop the option's life into many short steps, let the price move up or down by a fixed factor each step, and value the option backwards from expiry, using the risk-neutral probabilities at each node. For an American option, compare the value of continuing with the value of exercising at every node and keep the larger. As the number of steps grows, the tree converges to the continuous-time price.

Formally. With step Δt: $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, $p = (e^{r\Delta t} - d)/(u - d)$, and at each node $V = \max\big(\text{exercise value},\ e^{-r\Delta t}\,[pV_u + (1-p)V_d]\big)$.

Why it appears here. §9.2 (inverting American prices) and §9.4.

Deeper. Hull (2021); Dealer Hedging, section 3.2, for the one-step replication argument the tree repeats.

A.30 Delta conventions, and delta as a strike coordinate

The idea. Markets often name a strike by its delta rather than its price: the "25-delta put" is the put whose delta is -0.25. Because delta depends on the volatility at that strike, finding the strike for a given delta means solving an equation, and the answer moves when volatility moves. There are several deltas in use — spot or forward, and in foreign exchange "premium-adjusted" deltas that account for the premium being paid in the foreign currency — so a quoted "25 delta" needs its convention stated.

Formally. Forward call delta $N(d_1)$; spot delta multiplies by $e^{-q\tau}$; the premium-adjusted forward call delta is $(K/F)\,N(d_2)$. The strike for a target delta solves $N\big(d_1(K, \sigma_i(K))\big) = \Delta$. The delta-neutral straddle strike has $d_1 = 0$, that is $k = \frac{1}{2}\sigma_i^2\tau$.

Why it appears here. §10.3, §10.4, §11.1 and §11.3.

Deeper. Reiswich and Wystup (2010).

A.31 Look-ahead bias, and data snooping

The idea. Look-ahead bias is using information that was not available when a decision would have been made: an "end-of-day" implied volatility built from quotes after the close, a realised volatility aligned forward and then used as a signal, a surface refitted with tomorrow's data. Data snooping is testing many signals and reporting the best: with enough attempts something always looks significant. Both inflate backtests in exactly the option-implied signal literature of §13.6.

Why it appears here. §12.5, §13.6 and §15.4.

Deeper. Dealer Hedging, Appendix A.28.

A.32 Structural credit models

The idea. A firm's equity can be viewed as a call option on the firm's assets, struck at the face value of its debt: shareholders keep whatever is left after creditors are paid, and nothing if the assets fall short. In that picture the volatility of the firm's assets drives both its probability of default and its credit spread. Asset volatility cannot be observed, so it is backed out from the volatility of the equity — for which option-implied volatility is the forward-looking measure. This is the structural model Merton introduced in 1974.

Why it appears here. §13.7.

Deeper. Hull (2021), on credit risk.

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